InternSync – System Design Document

CSC 305 – Software Engineering · Semester 1 – 2026/2027 · Section 5MA1 · Group 3
Tasks T2 (System design) and T3 (UI design) · Milestone M2 · Version 1.7 · October 29, 2026

Version	Date	Description	Author
1.0	Oct 13, 2026	First version for Milestone M2: architecture, zero-cost deployment, data model (ERD and data dictionary), behaviour design, security design. The API specification is a separate document.	Salam Alsaour
1.1	Oct 13, 2026	Section 8: wireframes (task T3), their mapping to SRS Table 6.1, and the design system.	Salam Alsaour
1.2	Oct 20, 2026	CR-05: organizations (universities, colleges, companies), self-registration and verification, Platform Admin; companies publish vacancies directly; CSV import removed. Sections 1, 2, 4, 5, 6, 7, 8, 9, and 10 updated; ERD redrawn in four parts (49 tables).	Salam Alsaour
1.3	Oct 24, 2026	CR-07 (Increment 1 implementation): `PATCH /me` usable before consent; purged verification documents keep their file reference; jobs use transaction-level advisory locks; supervisors see the cycles of their placements; permission test described as built. Sections 2, 4, 5, and 7 updated.	Salam Alsaour
1.4	Oct 26, 2026	CR-08 (Increment 2 implementation): matching weights as three numeric settings; downloads always streamed by the API, so `files.signed_url_seconds` is removed; attendance rows may hold only an unlock; a final evaluation cannot be reopened while its grade is submitted or Final; row level security on every table; backup retention and GitHub Actions limits for a private repository. Sections 3.2, 3.4, 3.5, 4.3.2, 4.3.6, 5.1, and 5.5 updated.	Salam Alsaour
1.5	Oct 27, 2026	CR-09 (Increment 3 implementation): PDF exports with fpdf2 and HarfBuzz shaping instead of WeasyPrint; similarity passages of about one sentence, and a passage compared with the source report's nearby passages taken together; `slot_id` on passages and `attempts` on checks. Sections 1.3, 4.3.5, 5.6, and 6.3 updated.	Salam Alsaour
1.6	Oct 28, 2026	CR-10: offline mode with a saved copy of the session and drafts kept on the device (Sections 7.3 and 8.3). CR-11 (Increment 4 implementation): retention period per college, anonymization as built, `anonymized_at` on cycles and users, `purged_at` on files, the certificate page and public answer as built, the idle limit applied on the device. Sections 4.3.1, 4.3.2, 4.3.8, 4.5, 5.6, 5.7, 7.1, 7.3, and 8.3 updated.	Salam Alsaour
1.7	Oct 29, 2026	Increment 5: on phones the menu folds behind a button (Section 8.2), found while taking the user guide's screenshots.	Salam Alsaour

Related documents

- SRS v1.2: the requirements this design implements. Requirement IDs (FR-..., NFR-...) refer to it.
- API Specification: every endpoint, its roles, payloads, and error codes.
- Diagram pack: figures referenced below. The ERD figures are generated from `source/06_erd.py`, which must match Section 4.
- Change requests: decisions that changed the SRS or SPMP.

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1. Introduction

1.1 Purpose

This document describes how InternSync is built: its components, where they run, how data is stored, how the workflows behave, and how security and privacy requirements are met. It turns SRS v1.2 into decisions a developer can implement directly. The API contract is in the separate API Specification.

Since version 1.2 (CR-05), InternSync is one platform for many organizations: a Platform Admin run by the provider; universities, each with colleges; one Coordinator per college with the college's academic supervisors and students; and companies, each with a Company Admin and optional further company supervisors.

1.2 Design Principles

Principle	What it means in the design	Requirements
One deployable backend	A single FastAPI container holds the REST API, the AI services, and the scheduler. No message broker or separate worker.	C-03, C-10
Server-side authorization	Every endpoint declares its roles and record scope; queries are filtered by scope, never by the client.	FR-AUT-05
Organization isolation	Every query on academic records is filtered by the caller's college, and every query on company records by the caller's company. The Platform Admin has no endpoint that returns academic records.	FR-AUT-05, NFR-PRV-03
Explicit state machines	Vacancies, applications, report slots, evaluations, grades, and certificates change state only through named transitions, each checked and audited.	FR-WFL-01, FR-AUD-01
AI is advisory	AI outputs are stored and displayed with their reason and model version; no AI output changes a status.	NFR-AI-01, FR-MAT-07, FR-SIM-03
Configurable defaults	Every value marked "default" in the SRS is a row in `system_settings` or a cycle field.	NFR-MNT-03
Bilingual by construction	All texts are keys translated in both languages; dates are stored in UTC and shown in Asia/Riyadh time.	FR-LOC-01, FR-LOC-02, NFR-L10N-02
Zero cost	Only free tiers and open-source software; the same containers run locally with Docker Compose.	C-04, NFR-PRT-02

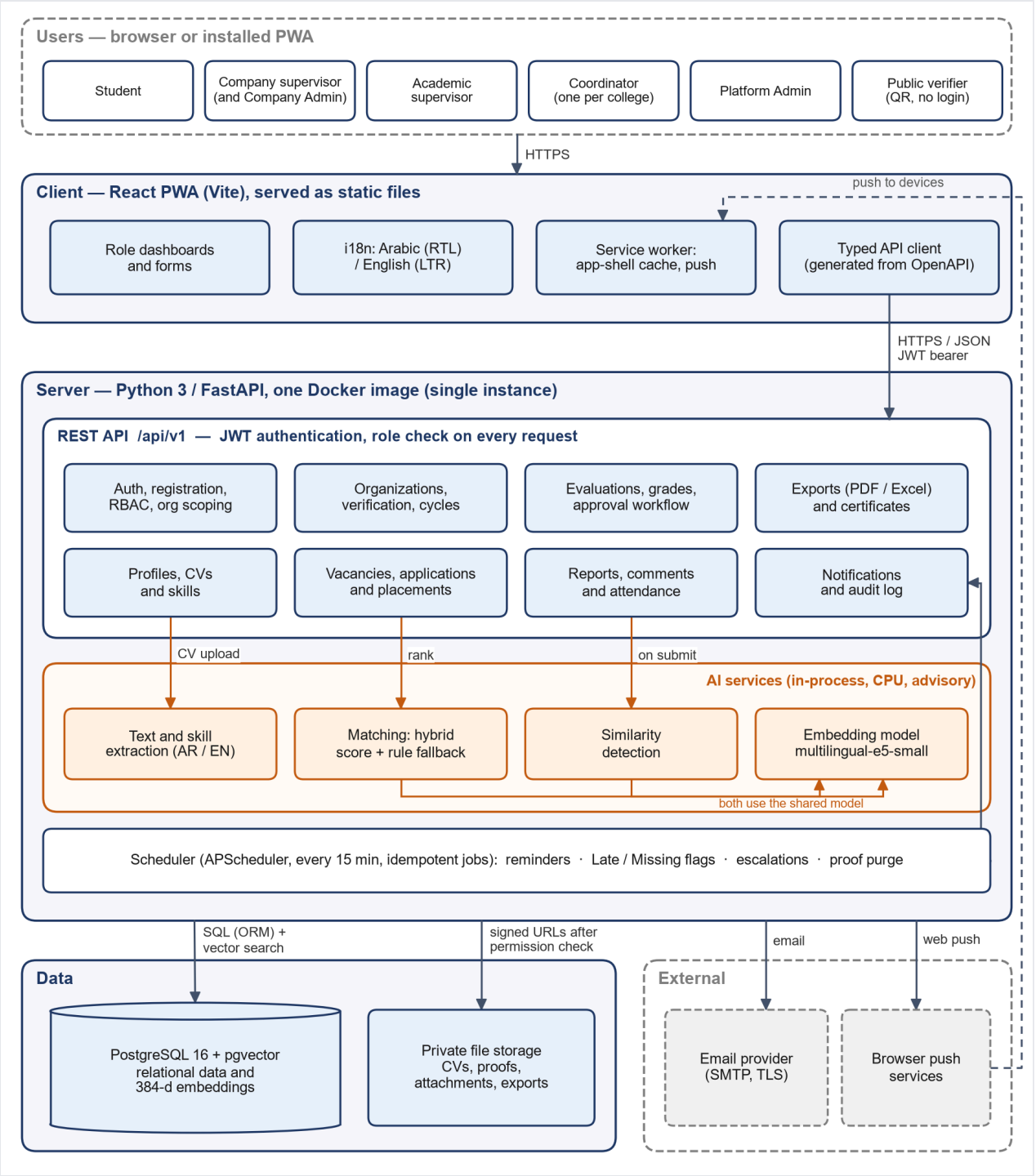
1.3 Technology Choices

Layer	Choice	Reason
Backend language and framework	Python 3.12, FastAPI, Pydantic v2	Required by C-03; generates the OpenAPI document (NFR-MNT-02).
Database access	SQLAlchemy 2.0 (synchronous), psycopg 3, Alembic migrations	Mature and simple; FastAPI runs synchronous endpoints in a thread pool, which is enough for the expected load (NFR-PER-01).
Database	PostgreSQL 16 or later with `pgvector` (`halfvec`) and `pg_trgm`	Relational data and embeddings in one place (C-03). `halfvec` halves vector storage to fit the free database size.

Layer	Choice	Reason
Passwords	Argon2id (`argon2-cffi`)	NFR-SEC-02.
Tokens	JWT (HS256, `PyJWT`) with a server-side session row	Idle timeout and logout revocation (FR-AUT-03).
Embedding model	`intfloat/multilingual-e5-small` (384 dimensions) with `sentence-transformers` on CPU	Arabic and English (NFR-L10N-04); small enough for CPU hosting (C-10).
Text extraction	`pypdf` for PDF, `python-docx` for DOCX	FR-PRF-04, FR-PRF-05.
File type check	Magic-byte detection (`filetype`) plus a DOCX structure check	NFR-SEC-04.
Scheduler	APScheduler 3 inside the API process, with Postgres advisory locks	NFR-REL-03.
PDF exports	fpdf2 with HarfBuzz text shaping (`uharfbuzz`) and bundled Noto Naskh Arabic and Noto Sans fonts (OFL)	Correct Arabic joining and RTL order (NFR-L10N-03, D-07); installs with pip on every platform, with no system libraries (CR-09).
Excel exports	`openpyxl`	FR-EXP-01.
QR codes	`qrcode`	FR-CER-01.
Email	Python `smtplib` over TLS, Jinja2 templates per language	FR-NOT-01, FR-NOT-06.
Web push	`pywebpush` with VAPID keys	FR-PWA-02.
Frontend	React 18, Vite, TypeScript, React Router, TanStack Query, react-i18next, Tailwind CSS (logical properties for RTL)	C-03, FR-LOC-01.
API client	`openapi-typescript` and `openapi-fetch`, generated from the server's OpenAPI document	NFR-MNT-02.
PWA	`vite-plugin-pwa` (Workbox)	FR-PWA-01 – 03.
Tests	pytest, pytest-cov, httpx test client; Vitest; Playwright smoke tests; Locust load tests	NFR-MNT-01, NFR-PER-01.

2. Architecture

2.1 Overview



The React PWA is a static build. It calls the REST API under `/api/v1` with a JWT bearer token. The API container holds:

- the **REST API modules** (Section 2.2);
- the **AI services**, in-process: text and skill extraction, matching, similarity detection, and one shared embedding model loaded once at start-up;
- the **scheduler**, which runs reminder, flag, escalation, and delivery jobs every few minutes.

PostgreSQL stores all records and embeddings. A private object store keeps CVs, attachments, exports, and certificates. Email goes out through SMTP and push messages through the browser vendors' push services.

2.2 Backend Modules

Each module is a Python package under `backend/app/` with the same files: `router.py` (endpoints), `schemas.py` (Pydantic models), `service.py` (business rules and transitions), `models.py` (SQLAlchemy tables), and `policy.py` (who may do what).

Module	Responsibility	Main tables	SRS
`core`	Settings, database session, error handling, i18n messages, audit helper, permission helpers, clock (UTC and Asia/Riyadh)	`system_settings`, `audit_log`	NFR-MNT-03, FR-AUD
`auth`	Registration, email confirmation, invitations, login, logout, sessions, password reset, lockout, consent	`users`, `user_sessions`, `account_tokens`	FR-AUT-02 – 04, FR-ORG-02, FR-ORG-06, NFR-SEC-02/03, NFR-PRV-01
`orgs`	Universities, colleges, companies, verification requests and documents, role transfer, account deactivation	`universities`, `colleges`, `companies`, `verification_requests`, `verification_documents`, `users`	FR-ORG-01 – 09, FR-AUT-06
`cycles`	Training cycles of a college, weeks, grade components, rubric, enrolments; skill vocabulary and majors (Platform Admin)	`training_cycles`, `cycle_weeks`, `grade_components`, `rubric_criteria`, `enrollments`, `skills`, `skill_resources`, `majors`	FR-CYC-01 – 05, FR-ORG-05
`profiles`	Student profile, CV upload, skill extraction and confirmation	`student_profiles`, `cvs`, `cv_extracted_skills`, `student_skills`	FR-PRF-02 – 05
`vacancies`	Vacancies published by companies, applications, placements	`vacancies`, `vacancy_skills`, `vacancy_majors`, `vacancy_universities`, `applications`, `placements`	FR-VAC-01 – 08
`matching`	Eligibility, fit score, explanation, development plan	`match_scores`	FR-MAT-01 – 07
`reports`	Report slots, versions, attachments, reviews, comments	`report_slots`, `report_versions`, `report_attachments`, `report_reviews`, `report_comments`	FR-REP-01 – 07, FR-WFL-01
`similarity`	Passages, similarity checks, supervisor decisions	`report_passages`, `similarity_checks`, `similarity_matches`	FR-SIM-01 – 03
`attendance`	Weekly attendance, unlocks	`attendance_weeks`	FR-ATT-01 – 04
`evaluations`	Periodic and final evaluations, student evaluation of the company, outcome survey	`evaluations`, `evaluation_scores`, `company_evaluations`, `outcome_surveys`	FR-EVL-01 – 05, FR-DSH-03
`grades`	Grade entry, computation, approval workflow	`grades`, `grade_component_scores`	FR-WFL-02 – 07
`dashboards`	Pending actions, cycle and platform statistics, leniency alerts	(read-only queries)	FR-DSH-01 – 05
`exports`	Grade sheet, term summary, training record	`files`	FR-EXP-01 – 03
`certificates`	Issue, revoke, public verification	`certificates`	FR-CER-01 – 03
`notifications`	In-app list, email and push delivery, push subscriptions	`notifications`, `push_subscriptions`	FR-NOT-01 – 06, FR-PWA-02

Module	Responsibility	Main tables	SRS
`files`	Upload checks, storage adapter (Supabase or local), permission-checked download	`files`	NFR-SEC-04
`jobs`	Scheduler set-up and job functions	`job_runs`	NFR-REL-03, FR-NOT-02 – 04
`ai`	Embedding model wrapper, text normalization, skill extractor, passage splitter	—	FR-PRF-04, FR-MAT, FR-SIM

2.3 Request Pipeline

1. **Transport:** HTTPS is terminated by the hosting provider (NFR-SEC-01). CORS allows only the configured frontend origins.
2. **Authentication:** the `current_user` dependency verifies the JWT signature and expiry, loads the session by its `sid`, rejects it if revoked or idle for more than the idle timeout, and updates `last_seen_at` (at most once per minute).
3. **Consent gate:** until `users.consent_at` is set, every endpoint except `GET /me`, `PATCH /me` (language, so the notice can be read in the other language), `POST /me/consent`, and `POST /auth/logout` returns `403 CONSENT_REQUIRED` (NFR-PRV-01).
4. **Verification gate:** an account whose verification is pending or rejected reaches only `/me`, its own verification requests, and, for students, the profile, CV, and vacancy list. Every other endpoint returns `403 VERIFICATION_REQUIRED` (FR-ORG-02).
5. **Authorization:** the route declares allowed roles; the service loads the record through a scope-filtered query that always includes the caller's college or company (Section 7.2). A record outside the caller's scope returns `403 FORBIDDEN`.
6. **Business rule and transition:** the service checks the state machine (Section 5) and limits, then writes the change.
7. **Audit and notifications:** the audit entry and any notifications are inserted in the **same database transaction** as the change, so either all are saved or none (FR-AUD-01, NFR-REL-04).
8. **Response:** JSON, or `application/problem+json` for errors with a code and a message in the user's language (NFR-USA-05).

2.4 Synchronous and Background Work

Work	How it runs	Limit
CV text and skill extraction	Synchronous in the upload request (thread pool)	≤ 30 s (NFR-PER-03)
Profile embedding	Synchronous when the student confirms skills	< 1 s
Vacancy embedding	Synchronous when a vacancy is published or edited	< 1 s
Ranking	Synchronous: one SQL query for eligible vacancies and their vectors, scoring in Python	≤ 5 s for 300 vacancies (NFR-PER-02)
Similarity check	After a report is submitted, a `similarity_checks` row is created as `pending` in the same transaction; a background task processes it right away, and the `similarity_retry` job picks up any check left pending	≤ 2 min (NFR-PER-04)
Email and push delivery	Notifications are written in the business transaction (outbox); the `deliver_notifications` job sends them	Every minute

Work	How it runs	Limit
Exports	Synchronous, streamed as a file	≤ 60 s for 100 students (NFR-PER-05)

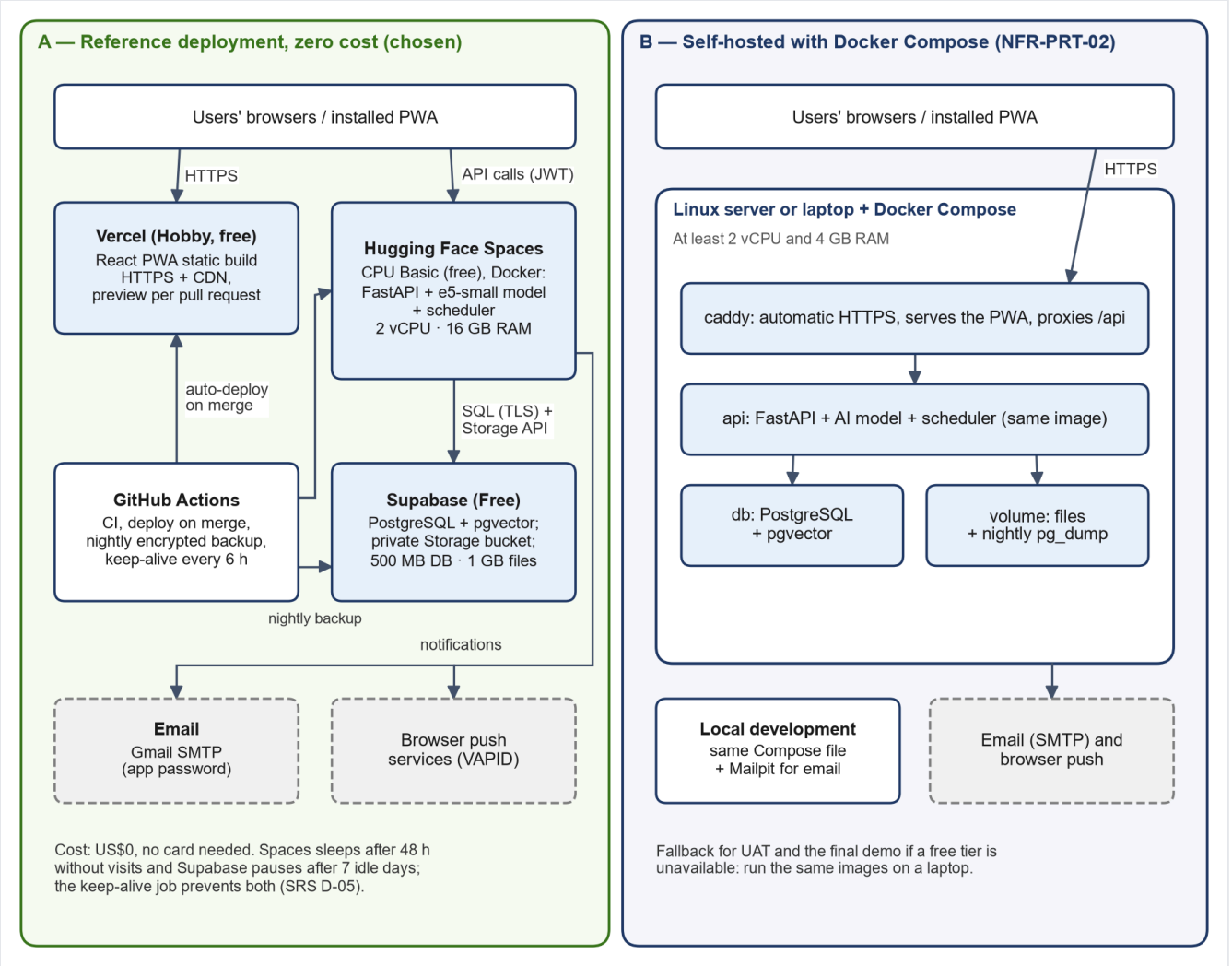
3. Deployment (zero cost)

3.1 Decision (CR-01)

The SRS v1.0 assumed one Linux VM, and the first diagram pack recommended Railway for the API. Both were replaced because the project must cost nothing (C-04):

- **Railway** is not free for this use. After a 30-day trial, the Free plan gives US$1 of credit per month and at most 0.5 GB RAM and 1 vCPU per service. The API with the embedding model needs about 1–1.5 GB of RAM and must run all the time for the scheduler. The Hobby plan costs US$5 per month.
- **Hugging Face Spaces** offers free CPU Basic hardware with 2 vCPU and 16 GB RAM for Docker containers, which fits the API, the model, and the scheduler with room to spare.

3.2 Reference Deployment



Component	Service and plan	Free-tier limits that matter	Notes
PWA (static build)	Vercel, Hobby	Personal, non-commercial projects	Auto-deploys from `main`; preview deployment per pull request; HTTPS and CDN.
API container (FastAPI, embedding model, scheduler)	Hugging Face Spaces, Docker SDK, CPU Basic	2 vCPU, 16 GB RAM, 50 GB disk that is **not persistent**; sleeps after 48 h without visits; must be a public Space	Port 7860. The container keeps no state on disk: data in Supabase, files in Supabase Storage. Secrets are stored as Space secrets.
Database	Supabase, Free	500 MB database (read-only above that); pauses after 7 days of low activity	PostgreSQL with `vector` and `pg_trgm`. The Data API is disabled, and row level security is on for every table as a second guard; the API connects as the owner of the tables through the session pooler over TLS.
File storage	Supabase Storage, Free (private bucket)	1 GB of files	Accessed only by the API with the service key; the API streams each download after the permission check, so no bucket link reaches a browser.
Email	Gmail SMTP with an app password	About 500 messages per day	A dedicated project Gmail account.
Push	Browser push services with VAPID keys	—	Payload holds the event title and a link only (NFR-PRV-03).
CI, backups, keep-alive	GitHub Actions	2,000 minutes a month and 500 MB of artifact storage for a private repository	See Section 3.5.

Why the limits are acceptable: the demonstration uses synthetic data for 2 universities and 4 colleges, about 400 students in total (C-09). With `halfvec(384)`, one embedding takes about 0.8 KB. 400 students × 13 reports × about 15 passages is about 78,000 passage vectors, about 60 MB plus about 80 MB of index, so the whole database stays well under 500 MB. Even the 10,000-report test of NFR-PER-04 (about 150,000 passages) fits in about 250 MB. Demo files are kept small (CVs about 100 KB) so storage stays under 1 GB.

3.3 Environments

Environment	What runs	How
Local development	`db` (pgvector/pgvector, PostgreSQL 17), `api` (FastAPI with hot reload), `web` (Vite dev server), `mailpit` (catches all email)	`docker compose up`
CI	PostgreSQL service container, backend tests with coverage, frontend lint, tests, and build, OpenAPI client check	GitHub Actions on every pull request
Reference (demo)	Vercel + Hugging Face Space + Supabase + Gmail SMTP	Deployed on merge to `main`
Self-hosted (NFR-PRT-02)	`caddy` (automatic HTTPS, serves the PWA, proxies `/api`), `api`, `db`, a file volume	`docker compose -f compose.prod.yml up -d` on any Linux server with at least 2 vCPU and 4 GB RAM

The storage adapter is chosen by `STORAGE_BACKEND=supabase|local`. Locally and when self-hosted, files live in a Docker volume outside any web root (NFR-SEC-04).

3.4 Configuration and Secrets

All configuration comes from environment variables. Secrets never enter the repository.

Variable	Purpose
`DATABASE_URL`	PostgreSQL connection string (TLS required in the reference deployment)
`JWT_SECRET`	Signing key for access tokens (32 random bytes or more); with `ENV=production` the API refuses to start without one
`FRONTEND_BASE_URL`, `CORS_ORIGINS`	Links in emails and QR codes; allowed browser origins
`STORAGE_BACKEND`, `SUPABASE_URL`, `SUPABASE_SERVICE_KEY`, `STORAGE_BUCKET`, `LOCAL_STORAGE_DIR`	File storage
`SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASSWORD`, `MAIL_FROM`	Email
`VAPID_PUBLIC_KEY`, `VAPID_PRIVATE_KEY`, `VAPID_SUBJECT`	Web push
`EMBEDDING_MODEL`	Model name; default `intfloat/multilingual-e5-small`
`AI_ENABLED`	`false` forces the rule-based fallback (FR-MAT-04 test)
`SCHEDULER_ENABLED`	`false` in tests and in a second instance
`ENV`	`development`, `test`, or `production`; development-only routes and demo accounts are refused in production
`PORT`, `FORWARDED_ALLOW_IPS`	Listening port (7860 on the Space) and the proxies whose forwarded client address is trusted (read by uvicorn)

3.5 CI/CD, Backups, and Keep-alive

Workflow	Trigger	Steps
`ci.yml`	Every pull request and push	Backend: `ruff`, `pytest` with coverage gate of 70% on business-logic modules (NFR-MNT-01), permission-matrix completeness test. Frontend: `eslint`, `vitest`, `vite build`, i18n missing-key check (FR-LOC-02), and a check that the generated API client matches the server's OpenAPI document.
`deploy-api.yml`	Push to `main` that changes `backend/` (merged pull requests have passed CI), or by hand	Copies the settings and secrets into the Hugging Face Space and pushes the files the image is built from, using `HF_TOKEN`. The image already holds the embedding model. The container runs `alembic upgrade head` before starting `uvicorn`. Skipped while the secrets are missing (Deployment).
Vercel Git integration	Push to `main`; pull requests	Builds and deploys `frontend/`.
`backup.yml`	Nightly at 02:00 Asia/Riyadh	`pg_dump` of the database plus a download of the storage bucket, each encrypted with a symmetric key from GitHub secrets and kept as a workflow artifact: the database for 30 days, the files for 7 days, to stay within the artifact storage (NFR-REL-02). A restore test is run once before M6.
`keepalive.yml`	Every 6 hours	Calls `GET /api/v1/health?deep=true`, which runs `SELECT 1` and reports the scheduler's last run. This keeps the Space awake and Supabase active (D-05) and serves as the uptime log (NFR-REL-01).

4. Data Design

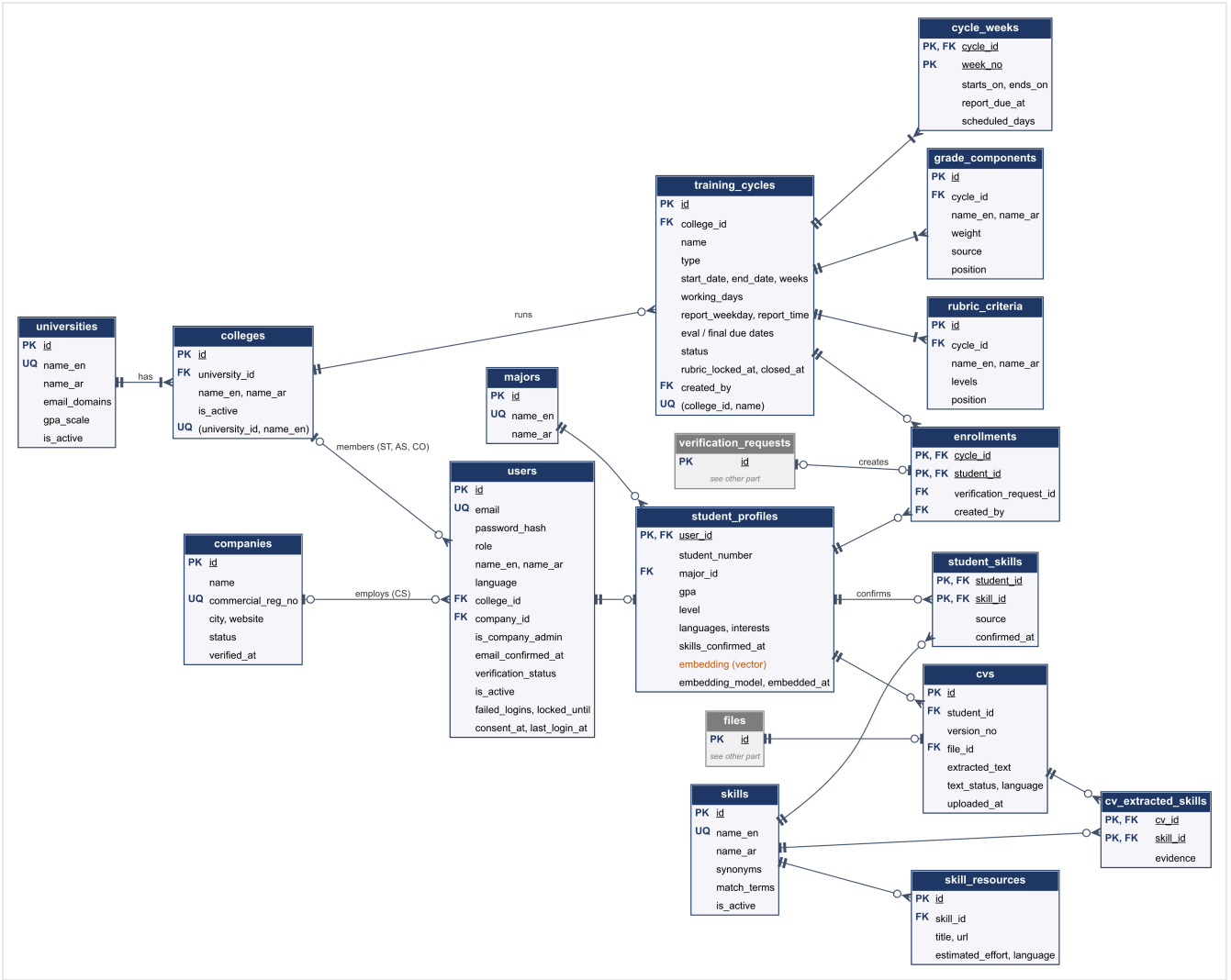
4.1 Conventions

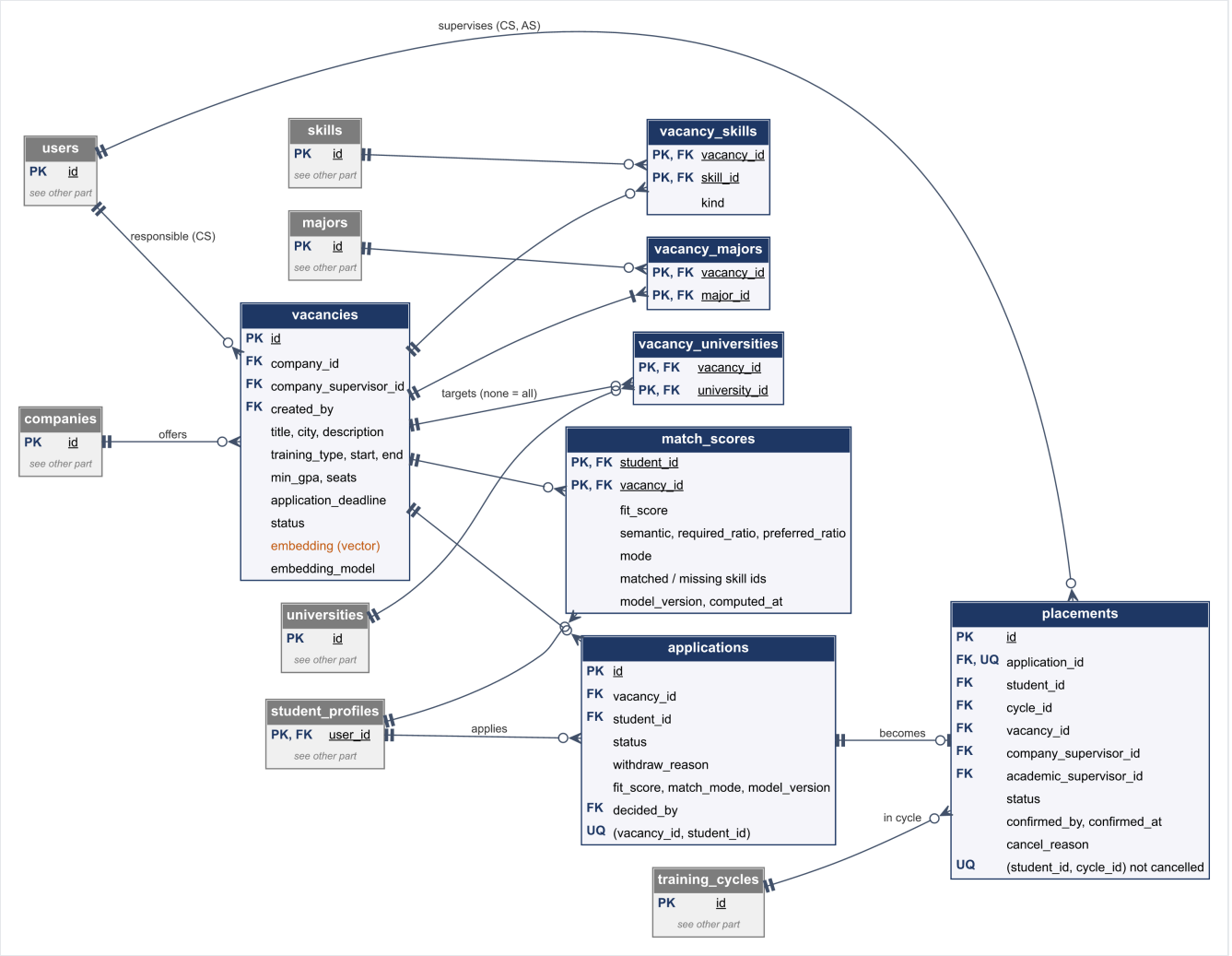
- Table names are plural `snake_case`. Primary keys are `id uuid default gen_random_uuid()`, except link tables (composite keys), `audit_log` and `report_passages` and `job_runs` (`bigint` identity), and `system_settings` (`key`).

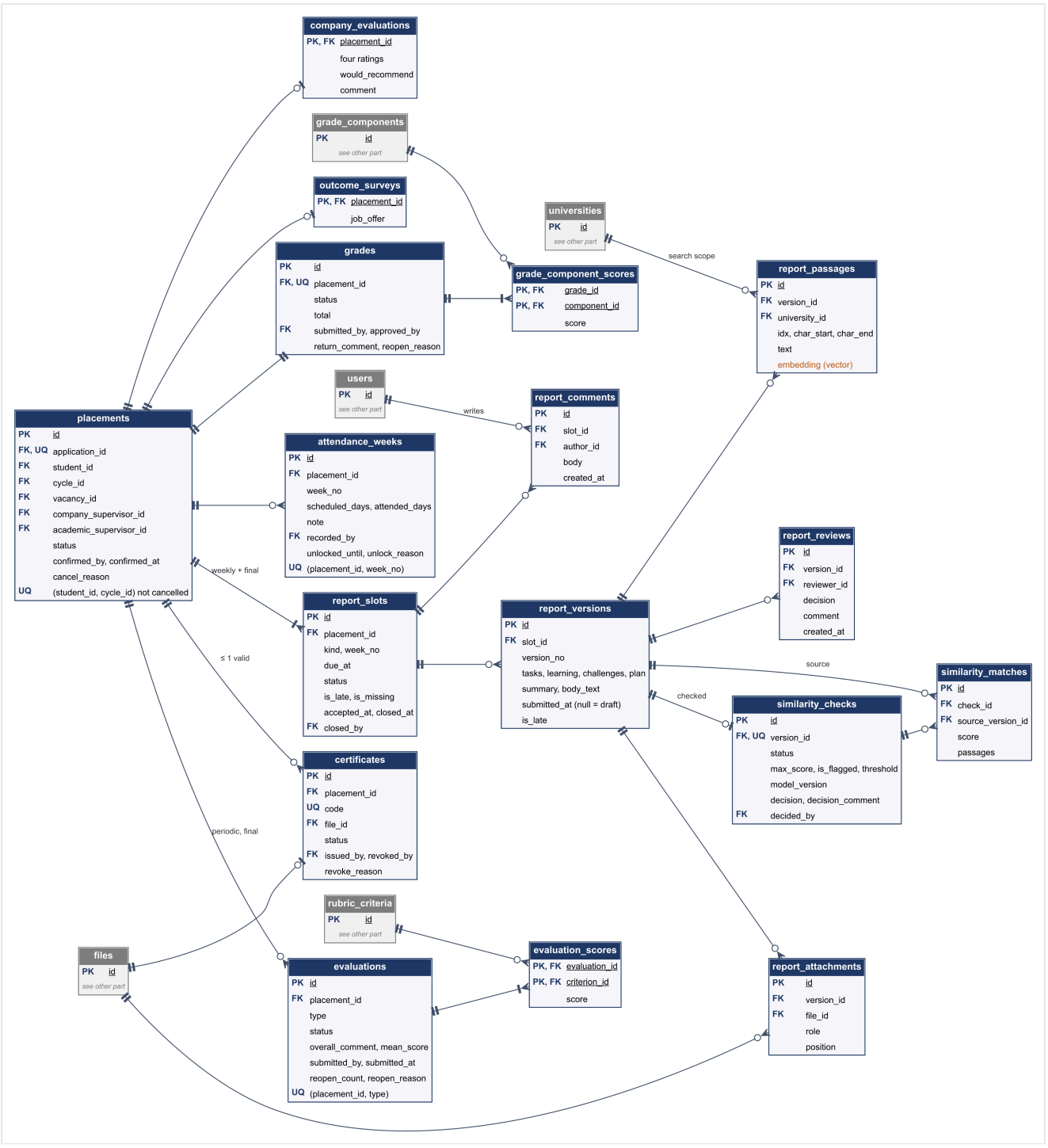
- Every timestamp is `timestamptz` stored in UTC. Deadlines are computed in Asia/Riyadh time and then stored in UTC (NFR-L10N-02).
- Status and type values are `text` with a `CHECK` constraint listing the allowed values (easier to migrate than PostgreSQL enums). The API uses the same lowercase values.
- Business records are **never deleted**: accounts are deactivated, placements cancelled, certificates revoked. Foreign keys use `ON DELETE RESTRICT`. Only technical rows (sessions, push subscriptions, job runs) may be deleted.
- Mutable tables have `created_at` and `updated_at`. Who changed what is kept in `audit_log`, not in extra columns, unless a column is needed for display or rules.
- Embeddings are `halfvec(384)` with an HNSW index using cosine distance.
- Text search on vacancies uses `ILIKE` with a `pg_trgm` GIN index, because PostgreSQL has no built-in Arabic dictionary.
- **Organization scope:** a student, an academic supervisor, and a coordinator carry `users.college_id`; a company supervisor carries `users.company_id`. Academic records reach their college through `placements.student_id` and their company through `vacancies.company_id`. Every scoped query joins to one of these columns (Section 7.2).

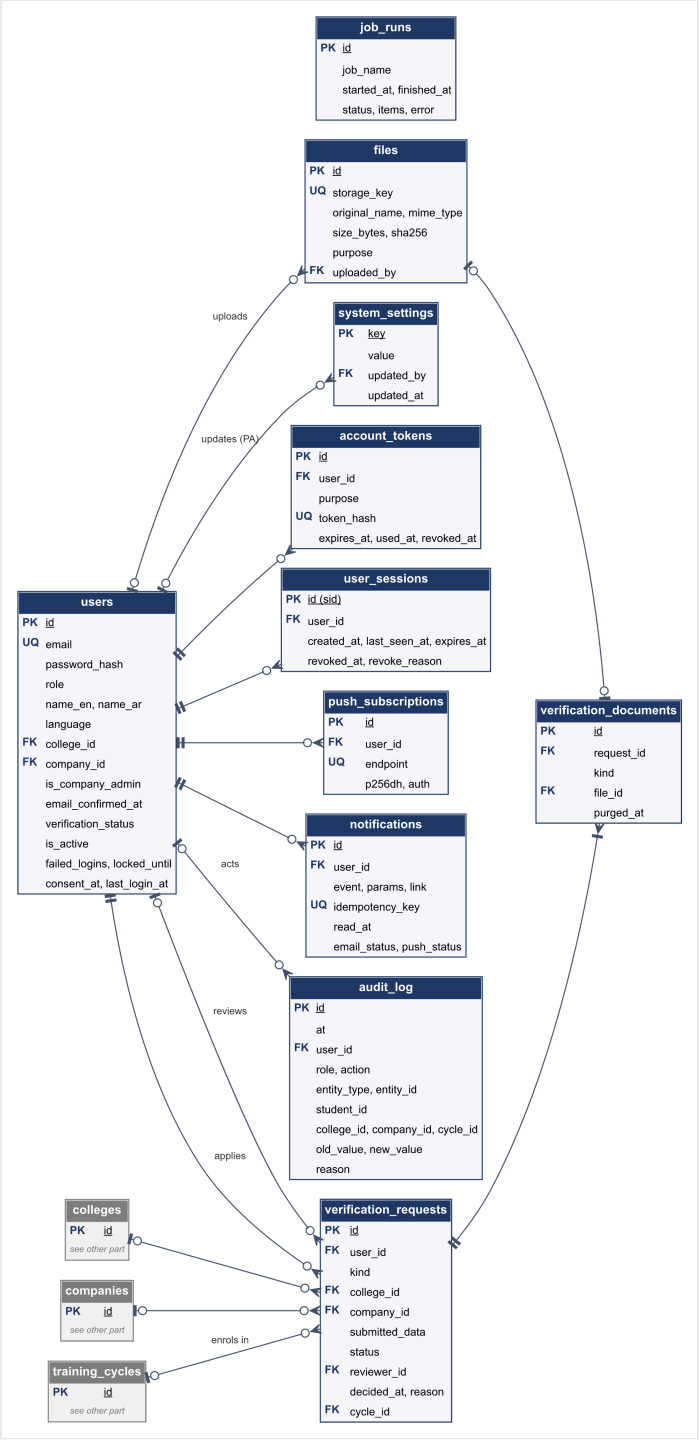
4.2 Entity-Relationship Diagrams

The data model has 49 tables in four diagrams. Crow's-foot notation; orange columns are vectors.




supervises (CS, AS)
users
PK id
see other part
responsible (CS)
skills
PK id
see other part
majors
PK id
see other part
vacancy_skills
PK, FK vacancy_id
PK, FK skill_id
kind
vacancy_majors
PK, FK vacancy_id
PK, FK major_id
vacancies
PK id
FK company_id
FK company_supervisor_id
FK created_by
title, city, description
training_type, start, end
min_gpa, seats
application_deadline
status
embedding (vector)
embedding_model
vacancy_universities
PK, FK vacancy_id
PK, FK university_id
targets (none = all)
companies
PK id
see other part
offers
match_scores
PK, FK student_id
PK, FK vacancy_id
fit_score
semantic, required_ratio, preferred_ratio
mode
matched / missing skill ids
model_version, computed_at
universities
PK id
see other part
applications
PK id
FK vacancy_id
FK student_id
status
withdraw_reason
fit_score, match_mode, model_version
FK decided_by
UQ (vacancy_id, student_id)
student_profiles
PK, FK user_id
see other part
applies
becomes
placements
PK id
FK, UQ application_id
FK student_id
FK cycle_id
FK vacancy_id
FK company_supervisor_id
FK academic_supervisor_id
status
confirmed_by, confirmed_at
cancel_reason
UQ (student_id, cycle_id) not cancelled
in cycle
training_cycles
PK id
see other part


company_evaluations
PK, FK placement_id
four ratings
would_recommend
comment
grade_components
PK id
see other part
outcome_surveys
PK, FK placement_id
job_offer
universities
PK id
see other part
search scope
grades
PK id
FK, UQ placement_id
status
total
FK submitted_by, approved_by
return_comment, reopen_reason
grade_component_scores
PK, FK grade_id
PK, FK component_id
score
report_passages
PK id
FK version_id
FK university_id
idx, char_start, char_end
text
embedding (vector)
users
PK id
see other part
writes
report_comments
PK id
FK slot_id
FK author_id
body
created_at
placements
PK id
FK, UQ application_id
FK student_id
FK cycle_id
FK vacancy_id
FK company_supervisor_id
FK academic_supervisor_id
status
confirmed_by, confirmed_at
cancel_reason
UQ (student_id, cycle_id) not cancelled
attendance_weeks
PK id
FK placement_id
week_no
scheduled_days, attended_days
note
FK recorded_by
unlocked_until, unlock_reason
UQ (placement_id, week_no)
weekly + final
report_reviews
PK id
FK version_id
FK reviewer_id
decision
comment
created_at
report_slots
PK id
FK placement_id
kind, week_no
due_at
status
is_late, is_missing
accepted_at, closed_at
FK closed_by
report_versions
PK id
FK slot_id
version_no
tasks, learning, challenges, plan
summary, body_text
submitted_at (null = draft)
is_late
source
similarity_matches
PK id
FK check_id
FK source_version_id
score
passages
checked
similarity_checks
PK id
FK, UQ version_id
status
max_score, is_flagged, threshold
model_version
decision, decision_comment
FK decided_by
≤ 1 valid
certificates
PK id
FK placement_id
UQ code
FK file_id
status
FK issued_by, revoked_by
revoke_reason
periodic, final
rubric_criteria
PK id
see other part
files
PK id
see other part
evaluations
PK id
FK placement_id
type
status
overall_comment, mean_score
submitted_by, submitted_at
reopen_count, reopen_reason
UQ (placement_id, type)
evaluation_scores
PK, FK evaluation_id
PK, FK criterion_id
score
report_attachments
PK id
FK version_id
FK file_id
role
position



Changes in version 1.2 (CR-05): new tables `universities`, `colleges`, `vacancy_universities`, `verification_requests`, and `verification_documents`; `import_batches` removed. `users` gained `college_id`, `is_company_admin`, `email_confirmed_at`, and `verification_status` (and lost `staff_id`, `department`, and `activated_at`); `companies` gained the commercial registration number and a status; `training_cycles` gained `college_id`; `vacancies` lost `cycle_id`, the `pending_review` status, and `return_comment`, and gained the training type and dates; `student_profiles.university_id` became `student_number`; `report_passages` gained `university_id`.

Changes from the ERD in diagram pack v1.0: the embedding moved from `CV` to `student_profiles` (the profile text includes confirmed skills and interests). New tables: `account_tokens`, `user_sessions`, `push_subscriptions`, `system_settings`, `cycle_weeks`, `majors`, `vacancy_majors`, `skill_resources`, `cv_extracted_skills`, `match_scores`, `files`, `report_reviews`, `report_passages`, `similarity_checks` (replaces the decision column of `SIMILARITY_FLAG`,

renamed `similarity_matches`), `import_batches`, and `job_runs`. `vacancies` gained `company_supervisor_id` and `return_comment`; `training_cycles` gained `working_days` and `rubric_locked_at`.

4.3 Data Dictionary

Abbreviations: **PK** primary key, **FK** foreign key, **UQ** unique, **NN** not null. Columns are nullable unless marked NN.

4.3.1 Organizations, accounts, and access

`universities` (FR-ORG-01).

Column	Type	Constraints	Notes
id	uuid	PK	
name_en, name_ar	text	NN, UQ on `lower(name_en)`	
email_domains	text[]	NN, default `{}`	e.g., `{example.edu.sa}`. A student email in one of these domains is shown to the Coordinator as a hint, not as proof
gpa_scale	numeric(3,2)	NN, default 5.00, CHECK in (4.00, 5.00)	Used to check student GPAs
is_active	boolean	NN, default true	
created_at	timestamptz	NN	

`colleges` (FR-ORG-01, FR-ORG-03). The college's single Coordinator is found in `users` (partial UQ below).

Column	Type	Constraints	Notes
id	uuid	PK	
university_id	uuid	NN, FK `universities`	
name_en, name_ar	text	NN; UQ (university_id, `lower(name_en)`)	
is_active	boolean	NN, default true	Inactive colleges are not offered at registration
retention_months	smallint	NN, default 12, CHECK 0–120	Months after a cycle closes before its personal data may be anonymized; set by the coordinator (NFR-PRV-04, CR-11)
created_at	timestamptz	NN	

`companies` (FR-ORG-04, FR-VAC-01).

Column	Type	Constraints	Notes
id	uuid	PK	
name	text	NN	As entered
commercial_reg_no	text	NN, UQ	Commercial registration number
city	text	NN	
website	text		Optional
status	text	NN, default `pending`, CHECK in (`pending`, `verified`, `rejected`, `suspended`)	Only a verified company can create vacancies or invite supervisors
verified_at	timestamptz		
created_at, updated_at	timestamptz	NN	

`users`: one row per account, exactly one role (FR-AUT-01).

Column	Type	Constraints	Notes
id	uuid	PK	
email	text	NN, UQ on `lower(email)`	Login name
password_hash	text		Argon2id; null until an invitation is accepted
role	text	NN, CHECK in (`student`, `company_supervisor`, `academic_supervisor`, `coordinator`, `platform_admin`)	
name_en, name_ar	text	NN	
language	text	NN, default `ar`, CHECK in (`ar`, `en`)	Interface, email, and notification language (FR-LOC-01, FR-NOT-06)
college_id	uuid	FK `colleges`; CHECK set if and only if role is `student`, `academic_supervisor`, or `coordinator`	
company_id	uuid	FK `companies`; CHECK set if and only if role is `company_supervisor`	
is_company_admin	boolean	NN, default false; CHECK true only for `company_supervisor`; partial UQ (company_id) where true	One Company Admin per company (FR-ORG-04)
email_confirmed_at	timestamptz		Set by the confirmation or invitation link (FR-AUT-02)
verification_status	text	NN, CHECK in (`pending`, `verified`, `rejected`, `not_required`)	`not_required` for invited users and the Platform Admin
is_active	boolean	NN, default true	FR-AUT-06
failed_logins	smallint	NN, default 0	NFR-SEC-03
locked_until	timestamptz		NFR-SEC-03
consent_at	timestamptz		NFR-PRV-01
last_login_at	timestamptz		
anonymized_at	timestamptz		Set by an anonymization; the account is inactive and its audit entries are masked (NFR-PRV-04, CR-11)
created_at, updated_at	timestamptz	NN	

Partial UQ (college_id) where role = `coordinator`, verification_status = `verified`, and is_active: one Coordinator per college (FR-ORG-03).

`account_tokens`: one-time email-confirmation, invitation, and password-reset links (FR-AUT-02, FR-AUT-04, FR-ORG-06).

Column	Type	Constraints	Notes
id	uuid	PK	
user_id	uuid	NN, FK `users`	
purpose	text	NN, CHECK in (`email_confirmation`, `invitation`, `password_reset`)	

Column	Type	Constraints	Notes
token_hash	text	NN, UQ	SHA-256 of the random token; the token itself is only in the email
expires_at	timestamptz	NN	72 h for confirmation and invitation, 60 min for reset (settings)
used_at	timestamptz		Set on use
revoked_at	timestamptz		Set when a newer token of the same purpose is issued
created_at	timestamptz	NN	

`verification_requests`: one request per registration attempt (FR-ORG-03 – 05, FR-ORG-09).

Column	Type	Constraints	Notes
id	uuid	PK	
user_id	uuid	NN, FK `users`	The applicant
kind	text	NN, CHECK in (`coordinator`, `company`, `student`)	
college_id	uuid	FK `colleges`	Coordinator and student requests
company_id	uuid	FK `companies`	Company requests
submitted_data	jsonb	NN	Data as submitted (student number, major, GPA, level; or company name, registration number, city), kept for history
status	text	NN, default `pending`, CHECK in (`pending`, `approved`, `rejected`)	Partial UQ (user_id) where `pending`: one open request per account
reviewer_id	uuid	FK `users`	Platform Admin for coordinators and companies; the college's Coordinator for students
decided_at	timestamptz		
reason	text	CHECK required when `rejected`	Shown to the applicant
cycle_id	uuid	FK `training_cycles`	Student approvals: the cycle the student is enrolled in
created_at	timestamptz	NN	

`verification_documents` (FR-ORG-08).

Column	Type	Constraints	Notes
id	uuid	PK	
request_id	uuid	NN, FK `verification_requests`	
kind	text	NN, CHECK in (`appointment_letter`, `commercial_registration`, `authorization_letter`, `course_registration`, `other`)	
file_id	uuid	FK `files`	Kept after the retention period: the `files` row stays as the record of what was submitted, while the stored object is deleted
purged_at	timestamptz		Set when the stored object is deleted; the file endpoint then answers `410 FILE_PURGED`

`user_sessions`: server side of each login, for idle timeout and logout (FR-AUT-03).

Column	Type	Constraints	Notes
id	uuid	PK	The `sid` claim in the JWT
user_id	uuid	NN, FK `users`	
created_at, last_seen_at	timestamptz	NN	Idle if `now - last_seen_at` > idle timeout
expires_at	timestamptz	NN	Absolute limit (12 h default)
revoked_at	timestamptz		
revoke_reason	text	CHECK in (`logout`, `idle`, `password_reset`, `deactivated`, `expired`)	
user_agent	text		For the user's own session list only

`push_subscriptions`: browser push endpoints (FR-PWA-02).

Column	Type	Constraints	Notes
id	uuid	PK	
user_id	uuid	NN, FK `users`	
endpoint	text	NN, UQ	Deleted when the push service answers 404 or 410
p256dh, auth	text	NN	Subscription keys
created_at, last_success_at	timestamptz		

4.3.2 Cycles and configuration

`training_cycles` (FR-CYC-01, FR-CYC-03, FR-CYC-05).

Column	Type	Constraints	Notes
id	uuid	PK	
college_id	uuid	NN, FK `colleges`	The college that runs the cycle
name	text	NN; UQ (college_id, name)	e.g., "Co-op Fall 2026"
type	text	NN, CHECK in (`coop`, `summer`)	
start_date, end_date	date	NN, CHECK end > start	
weeks	smallint	NN, CHECK 1–30	A-06; must fit between the dates
working_days	smallint[]	NN, default `{7,1,2,3,4}`	ISO weekdays (1 = Monday, 7 = Sunday); default Sunday–Thursday (CR-03, A-08)
report_weekday	smallint	NN, CHECK 1–7	Weekly report deadline day
report_time	time	NN	Local Asia/Riyadh time, e.g., 23:59
periodic_eval_due_at, final_eval_due_at, final_report_due_at	timestamptz	NN	
status	text	NN, CHECK in (`open`, `closed`)	Closed cycles are read-only (FR-CYC-05)
rubric_locked_at	timestamptz		Set when the first evaluation is submitted (FR-CYC-03)
closed_at	timestamptz		

Column	Type	Constraints	Notes
anonymized_at, anonymized_by	timestamptz, uuid	FK `users`	Set by the anonymization (Section 4.5); such a cycle cannot be reopened (CR-11)
created_by	uuid	NN, FK `users`	
created_at, updated_at	timestamptz	NN	

`cycle_weeks`: generated weeks and deadlines (FR-CYC-01, FR-REP-01, FR-ATT-01).

Column	Type	Constraints	Notes
cycle_id	uuid	PK, FK `training_cycles`	
week_no	smallint	PK, CHECK ≥ 1	
starts_on, ends_on	date	NN	Week k covers `start_date + 7(k−1)` to `start_date + 7k − 1`
report_due_at	timestamptz	NN	The `report_weekday` inside the week at `report_time` (Section 5.2); can be moved by the Coordinator
scheduled_days	smallint	NN, CHECK 0–7	Working days of the week inside the cycle dates

`grade_components` (FR-CYC-02, FR-WFL-02).

Column	Type	Constraints	Notes
id	uuid	PK	
cycle_id	uuid	NN, FK	
name_en, name_ar	text	NN; UQ (cycle_id, name_en)	
weight	numeric(5,2)	NN, CHECK > 0 and ≤ 100	The API replaces the whole list in one transaction and rejects a total other than 100
source	text	NN, CHECK in (`company_final_evaluation`, `academic_supervisor`)	At most one `company_final_evaluation` per cycle (partial UQ)
position	smallint	NN	Display order

`rubric_criteria` (FR-CYC-03, FR-EVL-01, FR-EVL-02).

Column	Type	Constraints	Notes
id	uuid	PK	
cycle_id	uuid	NN, FK	
name_en, name_ar	text	NN	
levels	jsonb	NN	`{"1": {"en": "…", "ar": "…"}, …, "5": {…}}`; all five levels required
position	smallint	NN; UQ (cycle_id, position)	

`skills`: the skill vocabulary, shared by all organizations and maintained by the Platform Admin (FR-CYC-04).

Column	Type	Constraints	Notes
id	uuid	PK	

Column	Type	Constraints	Notes
name_en	text	NN, UQ on `lower(name_en)`	Display name in English, e.g., "JavaScript"
name_ar	text	NN	
synonyms	text[]	NN, default `{}`	e.g., `{JS, جافا سكربت}`
match_terms	text[]	NN	Normalized forms of both names and all synonyms (Section 6.1); rebuilt on every edit
is_active	boolean	NN, default true	Inactive skills are not extracted
created_at, updated_at	timestamptz	NN	

`skill_resources` (FR-CYC-04, FR-MAT-06).

Column	Type	Constraints	Notes
id	uuid	PK	
skill_id	uuid	NN, FK `skills`	
title	text	NN	
url	text	NN, CHECK starts with `https://` or `http://`	
estimated_effort	text		e.g., "6 hours"
language	text	CHECK in (`ar`, `en`)	
position	smallint	NN	

`majors` (FR-PRF-02, FR-VAC-01).

Column	Type	Constraints	Notes
id	uuid	PK	
name_en	text	NN, UQ	Maintained by the Platform Admin; chosen by students at registration and by companies for eligibility
name_ar	text	NN	

`system_settings`: every SRS default, set by the Platform Admin for the whole platform (NFR-MNT-03).

Column	Type	Constraints	Notes
key	text	PK	Dotted name, see Table 4.1
value	jsonb	NN	Validated against a schema per key
updated_by	uuid	FK `users`	A Platform Admin
updated_at	timestamptz	NN	

Table 4.1 – Configurable defaults

Key	Default	Source
`auth.email_link_hours`	72	FR-AUT-02 (email confirmation and invitations)
`auth.reset_link_minutes`	60	FR-AUT-04
`auth.idle_timeout_minutes`	30	FR-AUT-03
`auth.session_max_hours`	12	Design
`auth.lockout_attempts` / `auth.lockout_minutes`	5 / 15	NFR-SEC-03

Key	Default	Source
`auth.password_min_length`	8	NFR-SEC-02
`applications.max_active`	5	FR-VAC-04
`files.cv_max_mb` / `files.attachment_max_mb`	5 / 10	FR-PRF-03, FR-REP-02
`reports.max_attachments`	3	FR-REP-02
`attendance.edit_days` / `attendance.pending_after_days`	7 / 3	FR-ATT-02, FR-ATT-04
`notifications.reminder_hours_before`	24	FR-NOT-02
`notifications.escalation_days`	3	FR-NOT-04
`matching.weight_semantic` / `matching.weight_required` / `matching.weight_preferred`	0.5 / 0.4 / 0.1	FR-MAT-02, Figure 8
`matching.semantic_floor` / `matching.semantic_ceiling`	0.70 / 0.95	Section 6.2 (tuned on the labelled set)
`similarity.threshold`	0.30	FR-SIM-02
`similarity.passage_cosine` / `similarity.ngram_size` / `similarity.ngram_containment`	0.90 / 5 / 0.5	Section 6.3
`dashboard.leniency_min_evaluations` / `dashboard.leniency_delta`	5 / 0.8	FR-DSH-04
`verification.document_retention_days`	30	FR-ORG-08
`auth.registrations_per_ip_hour`	5	Section 7.1

4.3.3 Students and CVs

`student_profiles` (FR-PRF-02).

Column	Type	Constraints	Notes
user_id	uuid	PK, FK `users`	Role must be `student`
student_number	text	NN, CHECK length 4–20	Entered at registration; unique within the university (checked by the service); read-only after verification
major_id	uuid	NN, FK `majors`	Entered at registration; read-only after verification
gpa	numeric(3,2)	NN, CHECK 0–5	Entered at registration; must not exceed the university's `gpa_scale`; read-only after verification
level	smallint		
languages, interests	text[]	NN, default `{}`	Editable by the student
skills_confirmed_at	timestamptz		Matching needs confirmed skills (FR-MAT-01)
embedding	halfvec(384)		Profile text embedding (Section 6.2)
embedding_model	text		NFR-AI-02
embedded_at	timestamptz		
updated_at	timestamptz	NN	

`enrollments`: which verified students belong to which cycle of their college (FR-ORG-05, FR-VAC-04).

Column	Type	Constraints	Notes
cycle_id	uuid	PK, FK	The cycle must belong to the student's college
student_id	uuid	PK, FK `users`	

Column	Type	Constraints	Notes
verification_request_id	uuid	FK `verification_requests`	The approval that created it
created_by	uuid	NN, FK `users`	The Coordinator
created_at	timestamptz	NN	

`cvs` (FR-PRF-03 – 05).

Column	Type	Constraints	Notes
id	uuid	PK	
student_id	uuid	NN, FK `users`	
version_no	smallint	NN; UQ (student_id, version_no)	The highest version is the current CV
file_id	uuid	NN, FK `files`	
extracted_text	text		
text_status	text	NN, CHECK in (`ok`, `no_text`, `failed`)	`no_text` triggers the FR-PRF-05 message
language	text	CHECK in (`ar`, `en`, `mixed`)	Detected from the script of the letters
uploaded_at	timestamptz	NN	

`cv_extracted_skills`: skills the extractor proposed for a CV (FR-PRF-04); kept for the evaluation script (NFR-AI-03).

Column	Type	Constraints	Notes
cv_id	uuid	PK, FK `cvs`	
skill_id	uuid	PK, FK `skills`	
evidence	text		The matched text (≤ 120 characters) shown to the student

`student_skills`: confirmed skills only (FR-PRF-02, FR-PRF-04).

Column	Type	Constraints	Notes
student_id	uuid	PK, FK `users`	
skill_id	uuid	PK, FK `skills`	
source	text	NN, CHECK in (`extracted`, `manual`)	
confirmed_at	timestamptz	NN	

4.3.4 Vacancies, applications, placements, matching

`vacancies` (FR-VAC-01 – 03, FR-VAC-07).

Column	Type	Constraints	Notes
id	uuid	PK	
company_id	uuid	NN, FK `companies`	The company must be `verified`
company_supervisor_id	uuid	NN, FK `users`	The supervisor responsible for the vacancy and its applicants ("own vacancies"); the creator by default; the Company Admin can change it
created_by	uuid	NN, FK `users`	
title, city, description	text	NN	GIN `pg_trgm` index on `title

Column	Type	Constraints	Notes
training_type	text	NN, CHECK in (`coop`, `summer`)	
training_start, training_end	date	NN, CHECK end > start	
min_gpa	numeric(3,2)		
seats	smallint	NN, CHECK > 0	
application_deadline	timestamptz	NN	
status	text	NN, CHECK in (`draft`, `published`, `closed`)	Section 5.1
published_at, closed_at	timestamptz		
embedding	halfvec(384)		Computed on publish and on edit
embedding_model	text		
created_at, updated_at	timestamptz	NN	

`vacancy_skills`: required and preferred skills.

Column	Type	Constraints	Notes
vacancy_id	uuid	PK, FK	
skill_id	uuid	PK, FK	
kind	text	NN, CHECK in (`required`, `preferred`)	

`vacancy_majors`: eligible majors (at least one).

Column	Type	Constraints	Notes
vacancy_id	uuid	PK, FK	
major_id	uuid	PK, FK	

`vacancy_universities`: target universities (FR-VAC-02). No row means the vacancy is open to every university.

Column	Type	Constraints	Notes
vacancy_id	uuid	PK, FK	
university_id	uuid	PK, FK `universities`	

`applications` (FR-VAC-04 – 07, FR-MAT-07).

Column	Type	Constraints	Notes
id	uuid	PK	
vacancy_id	uuid	NN, FK; UQ (vacancy_id, student_id)	No duplicates (FR-VAC-04)
student_id	uuid	NN, FK `users`	
status	text	NN, CHECK in (`submitted`, `accepted`, `declined`, `withdrawn`, `placed`, `placement_cancelled`)	Active = `submitted` or `accepted`
withdraw_reason	text	CHECK in (`student`, `placed_elsewhere`, `vacancy_closed`)	
fit_score	numeric(5,2)		Snapshot at application time
match_mode	text	CHECK in (`semantic`, `basic`)	

Column	Type	Constraints	Notes
model_version	text		NFR-AI-02
decided_by	uuid	FK `users`	
decided_at, withdrawn_at	timestamptz		
created_at, updated_at	timestamptz	NN	

`placements` (FR-VAC-07, FR-VAC-08).

Column	Type	Constraints	Notes
id	uuid	PK	
application_id	uuid	NN, FK, UQ	
student_id	uuid	NN, FK `users`	Partial UQ (student_id, cycle_id) where status ≠ `cancelled`
cycle_id	uuid	NN, FK	The student's enrolment cycle
vacancy_id	uuid	NN, FK	
company_supervisor_id	uuid	NN, FK `users`	A supervisor of the vacancy's company
academic_supervisor_id	uuid	NN, FK `users`	An academic supervisor of the student's college; can be changed with a reason (FR-VAC-08)
status	text	NN, CHECK in (`active`, `completed`, `cancelled`)	`completed` when the grade becomes Final
confirmed_by	uuid	NN, FK `users`	
confirmed_at	timestamptz	NN	
cancelled_at	timestamptz		
cancel_reason	text	CHECK required when cancelled	

`match_scores`: last computed fit per student and vacancy (FR-MAT-01 – 05, NFR-AI-02).

Column	Type	Constraints	Notes
student_id	uuid	PK, FK `users`	
vacancy_id	uuid	PK, FK	
fit_score	numeric(5,2)	NN	0–100
semantic, required_ratio, preferred_ratio	numeric(5,4)		Components (Section 6.2)
mode	text	NN, CHECK in (`semantic`, `basic`)	`basic` = "Basic matching" (FR-MAT-04)
matched_skill_ids, missing_skill_ids	uuid[]	NN	Explanation (FR-MAT-05)
model_version	text		
computed_at	timestamptz	NN	Recomputed when older than the profile's `embedded_at` or the vacancy's `updated_at`

4.3.5 Reports and similarity

`report_slots` (FR-REP-01, FR-REP-04, FR-REP-07, FR-WFL-01).

Column	Type	Constraints	Notes
id	uuid	PK	
placement_id	uuid	NN, FK	
kind	text	NN, CHECK in (`weekly`, `final`)	
week_no	smallint	CHECK set if and only if kind is `weekly`; UQ NULLS NOT DISTINCT (placement_id, kind, week_no)	
due_at	timestamptz	NN	Copied from `cycle_weeks` or `final_report_due_at`
status	text	NN, default `draft`, CHECK in (`draft`, `submitted`, `revision_requested`, `accepted`, `closed_missing`)	Figure 4.1
is_late	boolean	NN, default false	First submission after the deadline
is_missing	boolean	NN, default false	No submission at the deadline; stays true after a late submission
missing_flagged_at	timestamptz		
accepted_at, closed_at	timestamptz		
closed_by	uuid	FK `users`	
updated_at	timestamptz	NN	Index (status, due_at) for the jobs

`report_versions` (FR-REP-02, FR-REP-03, FR-REP-05).

Column	Type	Constraints	Notes
id	uuid	PK	
slot_id	uuid	NN, FK; UQ (slot_id, version_no)	
version_no	smallint	NN	
tasks_performed, learning, challenges, next_week_plan	text		Required on submit for weekly reports
summary	text		Optional for the final report (the main document carries the content)
body_text	text		Plain text compared for similarity: the four fields, or the text extracted from the final report's main document
submitted_at	timestamptz		Null means draft; partial UQ (slot_id) where null, so one draft per slot
is_late	boolean		`submitted_at` after the deadline minute
created_at, updated_at	timestamptz	NN	

`report_attachments`.

Column	Type	Constraints	Notes
id	uuid	PK	
version_id	uuid	NN, FK	
file_id	uuid	NN, FK `files`	A resubmission copies the rows, so one file can belong to several versions
role	text	NN, CHECK in (`main`, `attachment`)	`main` only for the final report

Column	Type	Constraints	Notes
position	smallint	NN	

`report_reviews`: the supervisor's decision on each submitted version (FR-REP-05).

Column	Type	Constraints	Notes
id	uuid	PK	
version_id	uuid	NN, FK	
reviewer_id	uuid	NN, FK `users`	
decision	text	NN, CHECK in (`accepted`, `revision_requested`)	
comment	text	CHECK required when `revision_requested`	
created_at	timestamptz	NN	

`report_comments` (FR-REP-06).

Column	Type	Constraints	Notes
id	uuid	PK	
slot_id	uuid	NN, FK	
author_id	uuid	NN, FK `users`	Student or academic supervisor
body	text	NN, CHECK length 1–2000	
created_at	timestamptz	NN	

`report_passages`: passages of submitted versions (FR-SIM-01).

Column	Type	Constraints	Notes
id	bigint	PK (identity)	
version_id	uuid	NN, FK	Index
slot_id	uuid	NN, FK `report_slots`	Copied from the version, so the search leaves out every version of the same report without a join
university_id	uuid	NN, FK `universities`	Copied from the student's college so the similarity search stays inside one university (FR-SIM-01)
idx	smallint	NN	Order in the report
char_start, char_end	integer	NN	Position in `body_text`
text	text	NN	
embedding	halfvec(384)	NN	HNSW index, `halfvec_cosine_ops`

`similarity_checks`: one per submitted version, with the supervisor's decision (FR-SIM-01 – 03).

Column	Type	Constraints	Notes
id	uuid	PK	
version_id	uuid	NN, FK, UQ	
status	text	NN, CHECK in (`pending`, `done`, `failed`)	
max_score	numeric(5,4)		Highest score over all sources
is_flagged	boolean	NN, default false	`max_score` ≥ threshold
threshold	numeric(4,3)	NN	Threshold in force when checked

Column	Type	Constraints	Notes
model_version	text		NFR-AI-02
attempts	smallint	NN, default 0	The retry job stops after 3
created_at	timestamptz	NN	
started_at, finished_at	timestamptz		
error	text		
decision	text	CHECK in (`no_issue`, `overlap_confirmed`)	Advisory only (FR-SIM-03)
decision_comment	text	CHECK required with a decision	
decided_by	uuid	FK `users`	
decided_at	timestamptz		

`similarity_matches`: overlap with each source report (top 5 sources).

Column	Type	Constraints	Notes
id	uuid	PK	
check_id	uuid	NN, FK; UQ (check_id, source_version_id)	
source_version_id	uuid	NN, FK `report_versions`	Author shown only to the Coordinator (FR-SIM-02)
score	numeric(5,4)	NN	Overlapping characters ÷ report characters
passages	jsonb	NN	`[{"report": [start, end], "source": [start, end], "cosine": 0.94}]`

4.3.6 Attendance and evaluations

`attendance_weeks` (FR-ATT-01 – 04). A row exists once the week is recorded or unlocked; an ended week without recorded days is pending. A check keeps `attended_days`, `recorded_by`, and `recorded_at` all set or all null, so a row may hold only an unlock.

Column	Type	Constraints	Notes
id	uuid	PK	
placement_id	uuid	NN, FK; UQ (placement_id, week_no)	
week_no	smallint	NN	
scheduled_days	smallint	NN, CHECK 0–7	From `cycle_weeks`
attended_days	smallint	CHECK 0 ≤ attended ≤ scheduled	Null until recorded
note	text		Absence note
recorded_by	uuid	FK `users`	Null until recorded
recorded_at	timestamptz		Null until recorded
updated_at	timestamptz	NN	
unlocked_until	timestamptz		Set by the Coordinator (FR-ATT-02)
unlocked_by	uuid	FK `users`	
unlock_reason	text		

`evaluations` (FR-EVL-01 – 04).

Column	Type	Constraints	Notes
id	uuid	PK	
placement_id	uuid	NN, FK; UQ (placement_id, type)	
type	text	NN, CHECK in (`periodic`, `final`)	
status	text	NN, CHECK in (`draft`, `submitted`)	
overall_comment	text		Required on submit
mean_score	numeric(4,3)		Computed on submit
submitted_by	uuid	FK `users`	
submitted_at	timestamptz		
reopen_count	smallint	NN, default 0, CHECK ≤ 1	Reopened at most once (FR-EVL-04)
reopened_at, reopened_by, reopen_reason			
updated_at	timestamptz	NN	

`evaluation_scores`.

Column	Type	Constraints	Notes
evaluation_id	uuid	PK, FK	
criterion_id	uuid	PK, FK `rubric_criteria`	Must belong to the placement's cycle (FR-EVL-02)
score	smallint	NN, CHECK 1–5	

`company_evaluations`: the student's evaluation of the training entity (FR-EVL-05). Never visible to company supervisors.

Column	Type	Constraints	Notes
placement_id	uuid	PK, FK	
learning_opportunities, supervision_quality, work_environment, relevance_to_major	smallint	NN, CHECK 1–5	
would_recommend	boolean	NN	
comment	text		
submitted_at	timestamptz	NN	

`outcome_surveys` (FR-DSH-03).

Column	Type	Constraints	Notes
placement_id	uuid	PK, FK	
job_offer	text	NN, CHECK in (`yes`, `no`, `pending`)	
submitted_at, updated_at	timestamptz	NN	

4.3.7 Grades and certificates

`grades` (FR-WFL-02 – 07). Created in `draft` when the placement is confirmed.

Column	Type	Constraints	Notes
id	uuid	PK	

Column	Type	Constraints	Notes
placement_id	uuid	NN, FK, UQ	
status	text	NN, default `draft`, CHECK in (`draft`, `submitted`, `final`)	Figure 4.2
total	numeric(5,2)		Weighted sum (Section 5.4)
submitted_by, approved_by	uuid	FK `users`	
submitted_at, approved_at	timestamptz		
return_comment	text		Last return comment
reopen_reason	text		Last reopen reason
updated_at	timestamptz	NN	

`grade_component_scores`.

Column	Type	Constraints	Notes
grade_id	uuid	PK, FK	
component_id	uuid	PK, FK `grade_components`	
score	numeric(5,2)	NN, CHECK 0–100	Computed for the company-evaluation component, entered for the others
updated_at	timestamptz	NN	

`certificates` (FR-CER-01 – 03).

Column	Type	Constraints	Notes
id	uuid	PK	
placement_id	uuid	NN, FK; partial UQ where status = `valid`	A revoked certificate can be replaced by a new one
code	text	NN, UQ	128 random bits in base32 (26 characters); the public ID in the QR code
file_id	uuid	NN, FK `files`	
status	text	NN, CHECK in (`valid`, `revoked`)	
issued_by	uuid	NN, FK `users`	
issued_at	timestamptz	NN	
revoked_by	uuid	FK `users`	
revoked_at	timestamptz		
revoke_reason	text	CHECK required when revoked	

4.3.8 Platform services

`files` (NFR-SEC-04).

Column	Type	Constraints	Notes
id	uuid	PK	

Column	Type	Constraints	Notes
storage_key	text	NN, UQ	`{purpose}/{yyyy}/{uuid}`; never derived from the user's file name
original_name	text	NN	Shown on download
mime_type	text	NN	Detected from the content, not the extension
size_bytes	integer	NN	
sha256	char(64)	NN	
purpose	text	NN, CHECK in (`cv`, `report_document`, `report_attachment`, `verification_document`, `export`, `certificate`)	Verification documents are kept under their own storage prefix
uploaded_by	uuid	FK `users`	
created_at	timestamptz	NN	
purged_at	timestamptz		Removed from storage by an anonymization; the file answers `410 FILE_PURGED` (CR-11)

`notifications`: in-app list and delivery outbox (FR-NOT-01 – 06).

Column	Type	Constraints	Notes
id	uuid	PK	
user_id	uuid	NN, FK	Index (user_id, read_at)
event	text	NN	e.g., `report.reminder`, `grade.final` (Table 5.6)
params	jsonb	NN, default `{}`	Values for the text template
link	text		Frontend route
idempotency_key	text	UQ	Prevents duplicates (NFR-REL-03)
read_at	timestamptz		
email_status, push_status	text	NN, CHECK in (`pending`, `sent`, `failed`, `skipped`)	
emailed_at, pushed_at	timestamptz		
email_attempts	smallint	NN, default 0	Up to 5 retries
created_at	timestamptz	NN	

`audit_log`: append-only (FR-AUD-01 – 03).

Column	Type	Constraints	Notes
id	bigint	PK (identity)	
at	timestamptz	NN, default now()	Server time
user_id	uuid	FK `users`	Null for the scheduler
role	text	NN	The four roles or `system`
action	text	NN	e.g., `create`, `update`, `submit`, `approve`, `reopen`, `unlock`, `login`, `deactivate`
entity_type	text	NN	e.g., `vacancy`, `report_slot`, `grade`
entity_id	text	NN	Index (entity_type, entity_id)
student_id	uuid		The student the record is about, for filtering (FR-AUD-02)

Column	Type	Constraints	Notes
college_id, company_id, cycle_id	uuid		Organization of the record, for scoping the log (FR-AUD-02)
old_value, new_value	jsonb		Changed fields only
reason	text		Mandatory reasons (reopen, cancel, unlock)

A trigger raises an error on `UPDATE` and `DELETE`, and the application's database role has only `INSERT` and `SELECT` on this table.

`job_runs` (NFR-REL-03). Rows older than 30 days are deleted.

Column	Type	Constraints	Notes
id	bigint	PK (identity)	
job_name	text	NN	
started_at, finished_at	timestamptz		
status	text	CHECK in (`ok`, `error`)	
items	integer		Number of items handled
error	text		

4.4 Derived Values (not stored)

Value	Computed from	SRS
Active applications of a student	`applications` with status `submitted` or `accepted`	FR-VAC-04
Eligibility and its reason	Student verified and enrolled in an open cycle; vacancy published, before its deadline, and open to the student's university (`vacancy_universities`); `vacancy_majors`; `min_gpa`; `training_type` equal to the type of the student's cycle (CR-06)	FR-MAT-03, FR-VAC-02, FR-VAC-04
Attendance percentage	Σ attended_days ÷ Σ scheduled_days over recorded weeks	FR-ATT-03
Dashboard counts and cycle statistics	Live queries (Section 5.7)	FR-DSH-01, FR-DSH-02
History of a record	`audit_log` filtered by entity	FR-AUD-03

4.5 Data Retention and Anonymization

Records stay for the life of the system. NFR-PRV-04 (Could) is one Coordinator action on a closed cycle, built in Increment 4 (design, Section 7):

- **When:** once `closed_at` plus the college's `retention_months` has passed; the coordinator sees the counts and confirms.
- **Removed:** names and contact details of that cycle's students (replaced by placeholders), their CV files, `extracted_text`, skills, embeddings, report text and attachments, comments, and certificate files.
- **Kept:** statistics-bearing columns (statuses, scores, dates, majors, companies, survey answers), so the cycle's figures do not change.
- **Skipped:** a student who is still in another cycle that is open or within its period.
- **Audit log:** append-only and unchanged; when read, the entries of an anonymized student are shown without their values.

Verification documents are deleted from storage 30 days after the decision by the `purge_verification_documents` job; the request, its decision, and its reason stay (FR-ORG-08).

5. Behaviour Design

Every transition below is one service function. It checks the actor's role and scope, checks the current state, writes the new state, adds one audit entry, and creates the listed notifications, all in one transaction. A transition that is not listed returns `409 INVALID_TRANSITION` (FR-WFL-01).

5.1 State Machines

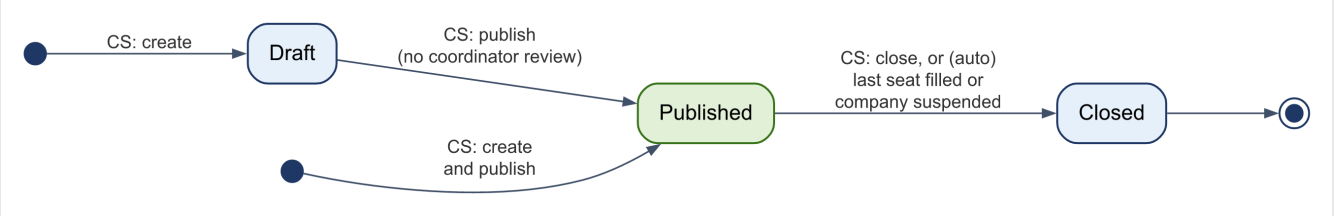


Table 5.1 – Vacancy transitions (FR-VAC-02, FR-VAC-07)

From	Action	Actor	Guard	To	Notifications
–	create	CS of a verified company	–	`draft`	–
–	create and publish	CS of a verified company	All required fields; deadline in the future	`published`	–
`draft`	publish	CS (own vacancy) or Company Admin	All required fields; deadline in the future	`published`	–
`published`	close	CS (own vacancy) or Company Admin	–	`closed`	–
`published`	(automatic)	system	Last seat filled by a placement, or the company suspended (Table 5.5)	`closed`	–

There is no review by a coordinator (CR-05); the coordinator's control is the placement confirmation (Table 5.2).

When a vacancy closes, its remaining active applications become `withdrawn` with reason `vacancy_closed` so they no longer count against the students' limit; the students are notified. This is a closure rule, not a decision on the applicant, so FR-MAT-07 still holds.

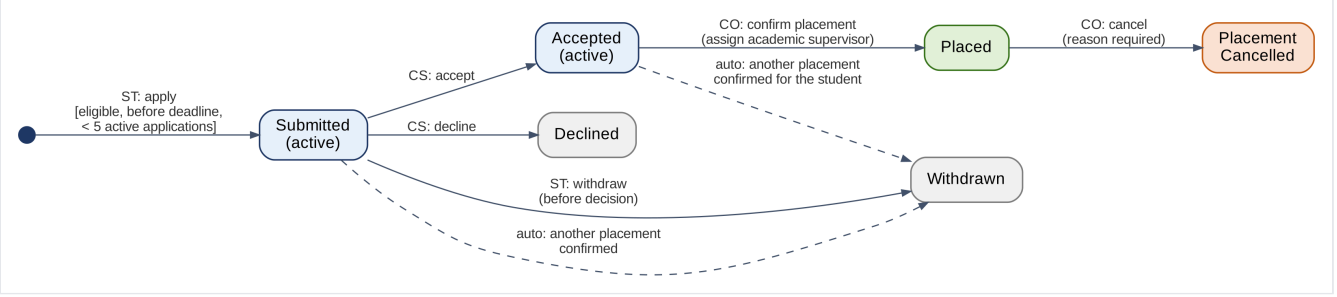


Table 5.2 – Application and placement transitions (FR-VAC-04 – 08)

From	Action	Actor	Guard	To	Notifications
–	apply	ST	Verified and enrolled in an open cycle; vacancy published, open to the student's university, and before its deadline; eligible; no earlier application; fewer than `applications.max_active` active	`submitted`	CS: new applicant
`submitted`	withdraw	ST	–	`withdrawn`	–
`submitted`	accept / decline	CS (own vacancy) or Company Admin	–	`accepted` / `declined`	ST: application decided; CO of the student: applicant accepted
`accepted`	confirm placement	CO of the student's college	No other non-cancelled placement for the student in the cycle; free seat; AS of the same college; CS of the vacancy's company	`placed`	ST, CS, AS: placement confirmed
`submitted`, `accepted`	(automatic)	system	Another placement of the student confirmed, or vacancy closed	`withdrawn`	ST
`placed`	cancel placement	CO	Reason required	`placement_cancelled`	ST, CS, AS

Confirming a placement also creates the report slots (Section 5.2), the grade in `draft`, and closes the vacancy if it was its last seat. Changing the academic supervisor is an update on the placement with a mandatory reason; it does not change the state.

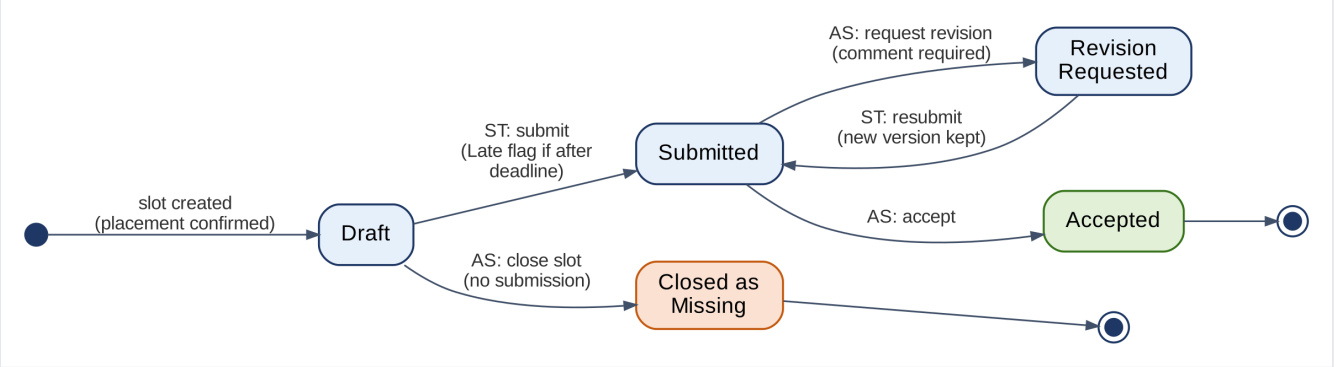


Table 5.3 – Report slot transitions (Figure 4.1; FR-REP-01 – 07, FR-WFL-01)

From	Action	Actor	Guard	To	Notifications
`draft`	submit	ST	Required fields; attachment limits; slot not closed	`submitted`	AS: report to review
`submitted`	accept	AS (assigned)	–	`accepted`	ST: report accepted
`submitted`	request revision	AS (assigned)	Comment required	`revision_requested`	ST: revision requested
`revision_requested`	resubmit	ST	As for submit	`submitted`	AS: report to review
`draft`	close as missing	AS (assigned)	No submitted version; deadline passed	`closed_missing`	ST

Saving a draft creates or updates the single draft version and does not change the slot's state. On resubmission the new draft starts as a copy of the last submitted version, including its attachments.

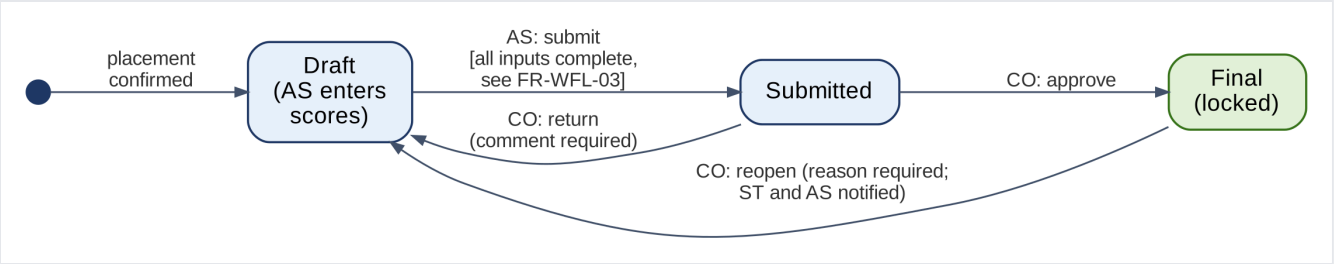


Table 5.4 – Grade transitions (Figure 4.2; FR-WFL-03 – 07)

From	Action	Actor	Guard	To	Notifications
`draft`	save scores	AS (assigned)	Scores 0–100	`draft`	–
`draft`	submit	AS (assigned)	Final evaluation submitted; final report `accepted`; every weekly slot `accepted` or `closed_missing`; every AS component entered. Otherwise `409 PRECONDITIONS_UNMET` with the list of missing items	`submitted`	CO: grade to approve
`submitted`	approve (single or bulk)	CO	–	`final`	ST, AS: grade final; placement becomes `completed`
`submitted`	return	CO	Comment required	`draft`	AS: grade returned
`final`	reopen	CO	Reason required	`draft`	ST, AS: grade reopened; placement back to `active`

Table 5.5 – Other transitions

Record	Transitions	SRS
Evaluation	`draft` → `submitted` (CS; every criterion rated, comment present; locks the cycle rubric); `submitted` → `draft` (CO reopen, reason required, once only; a final evaluation only while its grade is `draft`)	FR-EVL-01, FR-EVL-04, FR-CYC-03
Attendance week	record or edit (CS) while `now ≤ end of week + attendance.edit_days` or `now ≤ unlocked_until`; unlock (CO, reason required)	FR-ATT-01, FR-ATT-02
Certificate	issue (CO; grade `final`) → `valid`; revoke (CO; reason required) → `revoked`	FR-CER-01, FR-CER-03
Training cycle	`open` → `closed` (CO); `closed` → `open` (CO; reason required). While closed, every write to the cycle's records returns `409 CYCLE_CLOSED`	FR-CYC-05
Account	Self-registered: `pending` (email confirmed, request open) → `verified` or `rejected` with its request. active ↔ inactive (Platform Admin: any account; Coordinator: the college's academic supervisors and students; Company Admin: the company's supervisors). Deactivation revokes all sessions	FR-ORG-02, FR-AUT-06

Record	Transitions	SRS
Verification request	`pending` → `approved` or `rejected` (reason required). The Platform Admin decides coordinator and company requests; the college's Coordinator decides student requests and chooses the cycle. Approval sets the user to `verified`; for a company it also sets the company to `verified` and makes the representative its Company Admin; for a student it creates the enrolment. A rejected applicant may submit a new request	FR-ORG-03 – 05, FR-ORG-09
Company	`pending` → `verified` or `rejected` with its request; `verified` ↔ `suspended` (Platform Admin, reason required). Suspension closes the company's published vacancies and stops new ones; running placements continue	FR-ORG-04, FR-AUT-06
Coordinator role, Company Admin right	transfer to another verified member of the same college or company (Platform Admin, reason required); the previous holder keeps a normal account (an academic supervisor or a company supervisor)	FR-ORG-07

5.2 Deadlines and Time

- Week *k* of a cycle covers `start_date + 7(k−1)` to `start_date + 7k − 1`. Its report deadline is the day in that week whose ISO weekday equals `report_weekday`, at `report_time` in Asia/Riyadh, converted to UTC. The final report deadline is the cycle's `final_report_due_at`.
- `scheduled_days` of a week counts the cycle's working days that fall in the week and inside the cycle dates.
- **On time or late:** the deadline minute is inclusive. A submission is On time when `submitted_at < due_at + 60 seconds`, otherwise Late. With a 23:59 deadline, 23:58 and 23:59:30 are On time and 00:00 is Late (NFR-L10N-02).
- **Missing:** the `flag_missing` job marks a slot missing when `due_at` has passed and no version is submitted (FR-REP-04). It runs every 15 minutes and once 30 seconds after the API starts, so the flag appears within 15 minutes, well inside the 1-hour target, even after a restart.
- The Coordinator may move one week's deadline. The change updates the slots of that week that are still in `draft` and is audited.

5.3 Scheduled Jobs

All jobs run inside the API process under APScheduler. Each job runs in one transaction under a transaction-level PostgreSQL advisory lock (`pg_try_advisory_xact_lock`), so two running containers (for example during a redeploy) never run the same job twice, and a failed or crashed run never leaves the lock held. Jobs select work by **state**, not by time window, so a job missed during downtime catches up on its next run (NFR-REL-03).

Job	Every	Selects	Does	Idempotency key
`send_reminders`	15 min	Slots in `draft` or `revision_requested`, active placement, `now < due_at ≤ now + 24 h`	Notify the student (FR-NOT-02)	`reminder:{slot_id}`
`flag_missing`	15 min, and 30 s after the API starts	Slots in `draft`, `due_at < now`, `is_missing = false`	Set `is_missing`, notify the student and the AS (FR-REP-04, FR-NOT-03)	`missing:{slot_id}:{user_id}`
`escalate_overdue`	15 min	Reports not submitted, attendance weeks not recorded, evaluations not submitted, 3 or more days after their deadline	Notify the Coordinator (FR-NOT-04)	`escalation:{type}:{id}`
`similarity_retry`	5 min	`similarity_checks` still `pending` after 5 minutes, or `failed` fewer than 3 times	Run the check again	–

Job	Every	Selects	Does	Idempotency key
`deliver_notifications`	1 min	Notifications with email or push `pending`	Send email in the user's language and web push; retry failed email up to 5 times	–
`purge_verification_documents`	Daily	Documents of requests decided more than `verification.document_retention_days` ago	Delete the file from storage, set `purged_at` (FR-ORG-08)	–
`cleanup`	Daily	Expired sessions and tokens; `job_runs` older than 30 days	Delete technical rows	–

Table 5.6 – Notification events (FR-NOT-05)

Event	Recipient
`verification.submitted`	Platform Admin (coordinator and company requests); the college's CO (student requests)
`verification.approved`, `verification.rejected`	The applicant
`invitation.sent`	The invited person (email only)
`role.transferred`	The new and the previous holder
`application.received`	Vacancy's CS
`application.accepted`	CO of the student (placement to confirm)
`application.decided`, `application.withdrawn_auto`	ST
`placement.confirmed`, `placement.cancelled`, `placement.supervisor_changed`	ST, CS, AS
`report.reminder`, `report.revision_requested`, `report.accepted`	ST
`report.submitted`	AS
`report.missing`	ST, AS
`evaluation.submitted`	AS; ST for the periodic evaluation only (FR-EVL-03)
`grade.submitted`	CO
`grade.returned`	AS
`grade.final`, `grade.reopened`	ST, AS
`escalation.overdue`	CO
`certificate.issued`	ST

5.4 Grade Computation (FR-WFL-02)

- The company-evaluation component equals the mean of the final evaluation's criterion scores ÷ 5 × 100.
- The other components are entered by the academic supervisor (0–100).
- `total = Σ (weight × score) ÷ 100`, rounded half-up to two decimals. It is recomputed whenever a score changes and stored on submit and on approval.

5.5 File Handling (NFR-SEC-04, NFR-REL-04)

1. **Upload:** the request body is read in a stream with a size cap from the settings; a larger file is rejected with `413` before it is stored.
2. **Type check:** the type is detected from the first bytes. PDF (`%PDF-`), PNG, and JPEG are recognized by magic bytes. DOCX must be a ZIP containing `[Content_Types].xml` and `word/document.xml`. Anything else is rejected with `415`, so an executable renamed to `.pdf` is refused.
3. **Store:** the file is written to storage under a random key, then the `files` row is inserted.
4. **Report submission:** attachments are uploaded to the draft one by one. `submit` checks the limits and changes the state in one transaction, so a failed upload never leaves a half-submitted report.
5. **Download:** `GET /files/{id}` finds the record that owns the file, applies that record's permission rule, and streams the file from either storage (Supabase or local) with `Cache-Control: private, no-store` and `X-Content-Type-Options: nosniff`. No link to the bucket reaches a browser. Without permission it returns `403`.

5.6 Exports and Certificates (FR-EXP, FR-CER)

- **PDF:** drawn with fpdf2 and HarfBuzz text shaping. Noto Sans carries Latin text and Noto Naskh Arabic is its fallback, so a line can mix both. The page direction follows the user's language: in Arabic the text is right-aligned and table columns run from the right. Documents: grade sheet, term summary, training record, certificate.
- **Excel:** the grade sheet is built with `openpyxl`, one row per placement. Cell values starting with `=`, `+`, `-`, or `@` are prefixed with `'` to prevent formula injection.
- **Certificate:** one A4 landscape page in English and Arabic with the student name, university and college, cycle, company, training dates, the issue date, the certificate code, and a QR code for `https://<frontend>/verify/<code>`, without the grade. It is built once at issue and stored. The public page calls `GET /public/certificates/{code}` (the code is accepted with or without dashes, in any case) and shows Valid, Revoked, or Not found, without the grade (FR-CER-02). After an anonymization the page still answers, without the name (CR-11).

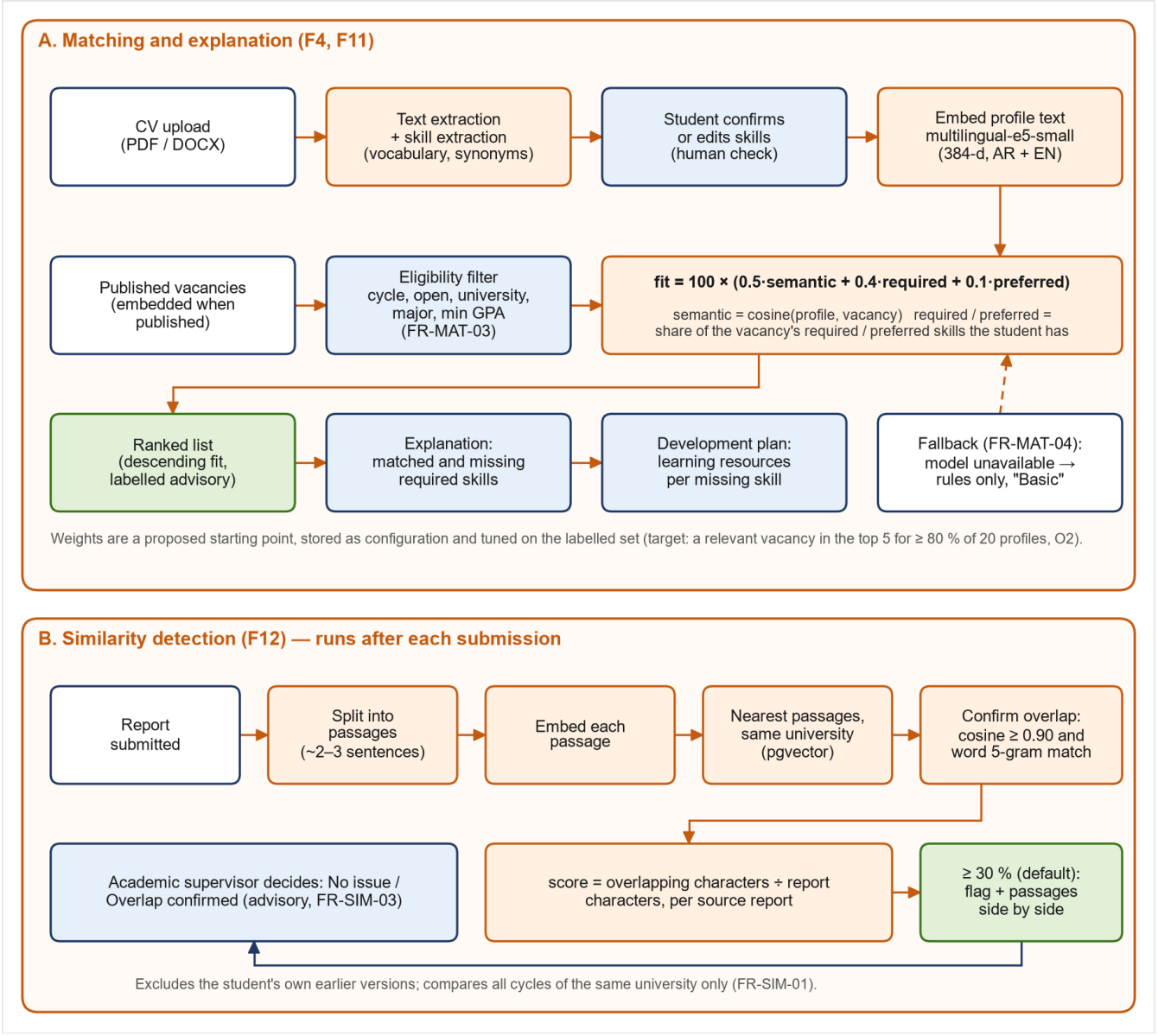
5.7 Dashboards (FR-DSH-01 – 04)

Counts are live queries, so they always equal the database (FR-DSH-01 acceptance).

Role	Pending actions
ST	Reports due in the next 7 days or overdue and not closed; revisions requested; applications by status; evaluation of the company and survey (from the training end date or the Final grade, whichever comes first)
CS	Applicants to review on own vacancies (the Company Admin: on all the company's vacancies); ended weeks without attendance (overdue after `attendance.pending_after_days`); evaluations due within 7 days or overdue
AS	Submitted reports not yet reviewed; flagged similarity checks without a decision; grades in `draft` whose preconditions are met
CO	Student verification requests of the college; accepted applications of the college's students without a placement; grades `submitted`; escalations (items overdue 3 days or more)
PA	Coordinator and company verification requests with their age; platform figures (FR-DSH-05)

Coordinator statistics per cycle and per company: placed and unplaced students, report submission rate and late rate, attendance rate, evaluation completion, and, if built, the job-offer rate (FR-DSH-03) and leniency alerts (FR-DSH-04: a company with at least 5 final evaluations whose mean is at least 0.8 above the cycle mean).

6. AI Services Design



The model `intfloat/multilingual-e5-small` is baked into the Docker image at build time, so the container never downloads it at run time and no text leaves the container (NFR-PRV-02). Every stored AI result records `embedding_model` or `model_version` as `<model name>@<revision>` (NFR-AI-02). If the model fails to load or `AI_ENABLED=false`, the services switch to their rule-based paths.

6.1 Text Normalization and Skill Extraction (FR-PRF-04, FR-CYC-04)

1. **Extract text** with `pypdf` or `python-docx`. If fewer than 50 letters come out, mark the CV `no_text` (FR-PRF-05).
2. **Normalize:** lowercase Latin letters; for Arabic, remove diacritics and tatweel, unify ه → ة ,ي → ى ,ا → آ/إ/أ; collapse spaces and punctuation.
3. **Match** the normalized text against every active skill's `match_terms` (both names and all synonyms, normalized the same way) as whole words or phrases of up to four words. Matching uses a pre-built automaton (Aho–Corasick) that is rebuilt when the vocabulary changes.
4. **Store** the matches in `cv_extracted_skills` with a short evidence snippet. The student then confirms, removes, or adds skills; only confirmed skills go to `student_skills`.

The method is rule-based on purpose: it is fast, explainable, and the student always confirms. Its recall is measured on the 20 labelled CVs (target ≥ 70%); missing synonyms are added to the vocabulary until the target is met.

6.2 Matching (FR-MAT-01 – 07)

1. **Eligible vacancies:** published, open to the student's university (no target universities, or the student's among them), before the deadline, the student's major among the eligible majors, GPA ≥ `min_gpa`, and the vacancy's training type equal to the type of the student's cycle (CR-06). The student must be verified and enrolled in an open cycle. Ineligible vacancies are left out of the ranking and show their reason when browsing (FR-MAT-03).
2. **Texts:** the profile text is the confirmed skills (both names), interests, and the start of the current CV text, cut to the model's 512-token limit and prefixed with `query: `. The vacancy text is title, description, and skill names, prefixed with `passage: `, as the e5 model expects.
3. **Score parts:**
   - `semantic = clamp((cos − floor) ÷ (ceiling − floor), 0, 1)`, with floor 0.70 and ceiling 0.95. This spreads the narrow cosine range of e5 over 0–1; both values are settings tuned on the labelled set.
   - `required = |S ∩ R| ÷ |R|` (1 when the vacancy lists no required skill).
   - `preferred = |S ∩ P| ÷ |P|`. When the vacancy has no preferred skills, the preferred weight moves to `required`.
4. **Fit score:** `fit = 100 × (0.5·semantic + 0.4·required + 0.1·preferred)`. Ties are ordered by the required part, then by the earlier deadline.
5. **Fallback:** when there is no profile or vacancy embedding, or the model is unavailable, `fit = 100 × (w_r·required + w_p·preferred) ÷ (w_r + w_p)` and the result is labelled "Basic matching" (FR-MAT-04).
6. **Explanation and plan:** matched skills are `S ∩ R`; missing skills are `R − S` (FR-MAT-05). The development plan lists each missing skill with its `skill_resources`, or "No resource available yet" (FR-MAT-06).
7. **Advisory:** the response carries `advisory: true`; no code path changes an application from a score (FR-MAT-07).

6.3 Similarity Detection (FR-SIM-01 – 03)

1. **Body text:** for weekly reports, the four text fields; for the final report, the text extracted from the main document.
2. **Passages:** split into sentences (on `.`, `!`, `?`, `؟`, `۔`, and line breaks); a sentence shorter than 8 words joins the next one, and one longer than 25 words is cut into pieces of 25 words, so a passage is about one sentence. Each passage is embedded with the `query: ` prefix and stored in `report_passages`. Passages of 40–80 words found only 4 of 10 reports that copied about half of their text: copied sentences shared a passage with the student's own words (CR-09).
3. **Candidates:** for each passage, take the 5 nearest passages by cosine (HNSW index) from submitted versions of **other** report slots, across all students and cycles of the same university; the search is filtered on `report_passages.university_id`, so reports of other universities are never compared (FR-SIM-01). Earlier versions of the same slot are excluded.
4. **Confirm:** a passage counts as overlap with a source report when one of that report's candidates reaches cosine ≥ 0.90 **and** at least 50% of the passage's word 5-grams (normalized text) appear in that report's candidates taken together, because the passages of two reports seldom line up. The lexical check stops two passages on the same topic from being flagged. The side-by-side view shows the candidates that share words with the passage.
5. **Score per source report:** characters of the checked report covered by confirmed passages ÷ total characters of the report.

6. **Flag:** when any source reaches the threshold (default 30%), the check is flagged and the top 5 sources are stored with their passage pairs for the side-by-side view (FR-SIM-02).
7. **Decision:** the academic supervisor records No issue or Overlap confirmed with a comment. Nothing else changes (FR-SIM-03).

The method is measured with the evaluation script on 10 copied and 10 original reports (target: at least 9 of 10 flagged, at most 1 false flag); the Increment 3 test report gives the results.

6.4 Evaluation Script (NFR-AI-03)

`ai-eval/` in the repository holds the labelled sets (`cvs/` with annotated skills, `profiles.json` with relevant vacancies, `reports/` with copied and original reports) and `evaluate.py`, which prints skill-extraction recall, top-5 hit rate (overall, Arabic, English), and similarity detection rates. The script uses the same service code as the API.

7. Security and Privacy Design

7.1 Authentication

- **Passwords:** Argon2id. At least 8 characters with letters and digits (NFR-SEC-02).
- **Lockout:** after 5 consecutive failures, `locked_until = now + 15 min`; during the lock every attempt, even with the right password, returns `423 ACCOUNT_LOCKED` (NFR-SEC-03).
- **Email-confirmation, invitation, and reset links:** 32 random bytes, URL-safe; only the SHA-256 hash is stored. Single use and time-limited. A password reset revokes all sessions of the user.
- **Registration:** at most `auth.registrations_per_ip_hour` registrations per IP address per hour. A new account cannot log in before its email is confirmed, and stays behind the verification gate (Section 2.3) until it is verified.
- **Forgot password** always answers `202`, whether or not the email exists, so accounts cannot be discovered.
- **Access token:** JWT HS256 with `sub`, `role`, `sid`, `iat`, and `exp` (12 h maximum). The session row enforces the 30-minute idle timeout and logout (FR-AUT-03). The app applies the same limit on the device, from the last key, click, or touch, online and offline; `GET /me` returns the limit (CR-11). The frontend keeps the token in memory and `sessionStorage` only (never in cookies, so there is no CSRF exposure), and Vercel sends a strict Content-Security-Policy to limit XSS.

7.2 Authorization (FR-AUT-05)

Each route declares its allowed roles. Record access then follows these scope rules, applied inside the database query:

Role	Scope rule
ST	`student_id = me` (profile, CVs, applications, placements and everything under them)
CS	Vacancies with `company_supervisor_id = me`, their applications and applicant profiles; placements with `company_supervisor_id = me`, their cycles, and their reports (read-only), attendance, and evaluations. The Company Admin also: every vacancy, application, and placement of the company, and the company's users. Never `company_evaluations`
AS	Placements with `academic_supervisor_id = me` (always students of the AS's college), their cycles, and their reports, similarity checks, evaluations, and grades. Source report authors are hidden
CO	Records whose student belongs to the CO's college (`users.college_id`), the college's cycles and users, and the student verification requests of the college. Applicant profiles of other colleges are never visible
PA	Universities, colleges, companies, users (account fields only), coordinator and company verification requests, skills, majors, settings, platform figures, and audit entries about accounts and organizations. No endpoint returns CVs, reports, evaluations, or grades to the PA

Role	Scope rule
Public	`POST /auth/register`, `GET /public/certificates/{code}`, `GET /health`, the lists of universities and colleges for the registration form, and the other auth endpoints that need no login

Permission tests: `backend/tests/permissions/matrix.yaml` lists every route with the callers allowed past the role gates (Public, an unverified student, ST, CS, CA, AS, CO, PA). One test calls every route as every caller that is not allowed and expects `401` or `403`, and a second test fails if a route registered in FastAPI is missing from the matrix, so every endpoint is covered for every role (FR-AUT-05 acceptance). Scope cases (own versus another user's record, and across organizations: a coordinator on another college's student, a company user on another company's vacancy) are tested in the module suites, whose test data hold two universities, three colleges, and two companies.

7.3 Data Protection

- **Verification documents:** stored under their own private prefix, downloadable only by the uploader and the reviewer, and deleted 30 days after the decision (FR-ORG-08).
- **Supabase:** the Data API (PostgREST) is disabled and Row Level Security is enabled with no policies on every table, so the public `anon` key cannot read anything. The API uses its own database role.
- **Audit log:** append-only by trigger and privileges (FR-AUD-02).
- **Push payloads:** event title and link only (NFR-PRV-03).
- **Logs:** request logs keep method, path, status, duration, and user ID, never request bodies, tokens, or CV text.
- **Responses per role:** shaped by role, for example the similarity source author only for the CO, and a grade for the student only when Final (FR-WFL-06).
- **Secrets:** stored as Space secrets, Vercel environment variables, and GitHub secrets.
- **Backups:** encrypted before upload (Section 3.5).
- **Consent:** a notice in the user's language at first login; `consent_at` is stored (NFR-PRV-01).
- **Offline copy on the device (CR-10):**
  - The saved copy of the pages shown is kept in the window's `sessionStorage` and deleted at sign-out, at the idle limit, and when the window closes.
  - Drafts typed offline are kept in `localStorage` under the user's ID until saved, for 7 days at most, and deleted at sign-out.
  - Files, exports, the audit log, and the Platform Admin's pages are never kept.
- **Anonymization:** Section 4.5 (NFR-PRV-04).
- **Release checks:** OWASP ZAP baseline of the web app and API scan in safe mode (`security-scan.yml`, weekly and on request), plus `pip-audit` and `npm audit` in CI (NFR-SEC-05).

8. User Interface Design

8.1 Wireframes (task T3)

The wireframes are a clickable prototype made in Claude Design: InternSync Wireframes (private until shared from the canvas's Share menu). Their sources are copied in `wireframes/`. The screens are in Arabic (RTL) by default; one screen repeats the student dashboard in English (LTR) to show the mirroring. In Play mode, the sidebar links move between screens. Some screens have tweaks that switch between the states a test must cover.

Table 8.1 – Wireframes against SRS Table 4.1

SRS screen	Artboards	States shown (tweaks)	Requirements
Registration, login, email confirmation, password reset	`Register`, `Login`, `Consent`	account type (student, coordinator, company); wrong password, account locked	FR-AUT-02 – 04, FR-ORG-02 – 05, NFR-SEC-03, NFR-PRV-01
Verification status and documents	`Verification_Pending`	pending; rejected with a new request	FR-ORG-02, FR-ORG-08, FR-ORG-09
Dashboard	`Main` (student), `ST_Dashboard_Mobile`, `ST_Dashboard_EN`, `CS_Dashboard`, `AS_Dashboard`, `CO_Dashboard`, `PA_Dashboard`	offline banner (student)	FR-DSH-01, FR-DSH-02, FR-DSH-05, FR-LOC-01, FR-PWA-03, NFR-USA-03
Profile and CV	`ST_Profile`	CV without readable text	FR-PRF-02 – 05
Vacancies (browse and ranked)	`ST_Vacancies`	semantic or basic matching	FR-VAC-03 – 05, FR-MAT-01 – 07
Vacancy editor and applicants	`CS_VacancyEditor`, `CS_Applicants`	company not yet verified	FR-VAC-01, FR-VAC-02, FR-VAC-06, FR-ORG-02
Company team	`CS_Team`	–	FR-ORG-04, FR-ORG-06, FR-AUT-06
Placements	`CO_Placements`	–	FR-VAC-07, FR-VAC-08
Report form and history	`ST_Report`, `AS_ReportReview`	similarity flagged or not	FR-REP-01 – 07, FR-SIM-02, FR-SIM-03
Attendance	`CS_Attendance`	–	FR-ATT-01 – 04
Evaluation form	`CS_Evaluation`	incomplete form with errors	FR-EVL-01 – 03
Grades	`AS_Grade`, `CO_Grades`	preconditions met or not	FR-WFL-02 – 07
Cycle settings	`CO_CycleSettings`	weights not totalling 100	FR-CYC-01 – 03
Student verification and academic supervisors	`CO_Students`, `CO_Supervisors`	approve and enrol; reject without a reason	FR-ORG-05, FR-ORG-06, FR-ORG-08, FR-AUT-06
Platform administration	`PA_Dashboard`, `PA_Verifications`, `PA_Organizations`, `PA_Settings`	company or coordinator request	FR-ORG-01, FR-ORG-03, FR-ORG-04, FR-ORG-07, FR-CYC-04, FR-DSH-05, NFR-MNT-03
Exports and certificates	`CO_Exports`	–	FR-EXP-01 – 03, FR-CER-01, FR-CER-03
Audit log	`CO_Audit`	–	FR-AUD-02
Certificate verification	`Verify_Mobile`	valid, revoked, not found	FR-CER-02

Not yet drawn: the student's evaluation of the company and the outcome survey (FR-EVL-05, FR-DSH-03, both Could), the notification list, and the Platform Admin's audit view (the same layout as `CO_Audit`, limited to accounts, organizations, and verification).

8.2 Design System

Token	Value	Use
Typeface	IBM Plex Sans Arabic (400, 500, 600, 700), one family for Arabic and Latin	All text
Primary	`#1F3864` (navy, as in the diagram pack); active item `#2E4D84`	Sidebar, primary buttons, headings of key figures

Token	Value	Use
Ground / surface / border	`#F4F6F9` / `#FFFFFF` / `#DCE2EA`	Page, cards, dividers
Text / muted text	`#1A2233` / `#566173`	Body, secondary text (6:1 contrast on white)
Advisory (AI)	text `#9A4410` on `#FFF4E8`	Every AI result, labelled "استرشادي" (Advisory)
Status chips	Draft `#4A5363/#EEF1F5`; Submitted `#1F4E9A/#E6EEFB`; Accepted, Final `#1E6B34/#E3F3E7`; Late `#8A4B00/#FFF1D6`; Missing, error `#A61B1B/#FCE8E8`; Revision requested `#6B3FA0/#F1EAFB`	Status text always shown with its colour
Shape	Cards 12 px radius; inputs and buttons 8 px; chips fully rounded	–
Touch targets	44 px minimum (48 px for main actions)	NFR-USA-04
Layout	Sidebar plus content in one wrapping row; on phones the menu folds behind a "Main menu" button above the content, so each page opens on its content	NFR-USA-03
Dates	`DD/MM/YYYY HH:mm`, Western digits	NFR-L10N-02

8.3 Rules for the Frontend

- **Routes by role**, after login:
  - ST: `/student` (dashboard), `/student/profile`, `/student/vacancies`, `/student/applications`, `/student/reports`.
  - CS: `/company` (dashboard), `/company/vacancies`, `/company/students/:placementId`, `/company/team` (Company Admin only).
  - AS: `/academic` (dashboard), `/academic/students/:placementId`, `/academic/grades`.
  - CO: `/coordinator` (dashboard), `/coordinator/students` (verification), `/coordinator/supervisors`, `/coordinator/cycles`, `/coordinator/placements`, `/coordinator/grades`, `/coordinator/exports`, `/coordinator/audit`.
  - PA: `/platform` (dashboard), `/platform/verifications`, `/platform/organizations`, `/platform/settings`, `/platform/audit`.
  - Public: `/register`, `/verify/:code`.
  - Shared: `/login`, `/confirm/:token`, `/invite/:token`, `/reset/:token`, `/verification` (own request status), `/notifications`.
- **Layout:** `dir` and `lang` on `<html>` follow the selected language. Spacing uses logical properties (`ms-`, `me-`, `ps-`, `pe-`), and directional icons are mirrored in Arabic.
- **Status chips:** text plus colour, never colour alone. AI results sit in a panel labelled "Advisory" (4.1 of the SRS).
- **Task length:** report submission, attendance, and evaluation each fit in at most 3 screens (NFR-USA-02); "Full attendance" is one button.
- **Offline:** the service worker caches the app shell. Offline, a banner is shown with the time of the saved copy, pages opened earlier in the session show that copy, and every submit button is disabled. Report and evaluation drafts typed offline are kept on the device and saved when the connection returns (FR-PWA-03, CR-10).

9. Traceability

SRS area	Module (2.2)	Tables (4.3)	API section
FR-AUT	`auth`, `orgs`	users, user_sessions, account_tokens	3.1, 3.2
FR-ORG	`orgs`, `auth`, `cycles`	universities, colleges, companies, users, verification_requests, verification_documents, enrollments	3.1, 3.2
FR-CYC	`cycles`	training_cycles, cycle_weeks, grade_components, rubric_criteria, skills, skill_resources, majors, system_settings	3.3, 3.4
FR-PRF	`profiles`	student_profiles, cvs, cv_extracted_skills, student_skills	3.5
FR-VAC	`vacancies`	vacancies, vacancy_skills, vacancy_majors, vacancy_universities, applications, placements	3.6, 3.7
FR-MAT	`matching`	match_scores	3.8
FR-REP, FR-WFL-01	`reports`	report_slots, report_versions, report_attachments, report_reviews, report_comments	3.9
FR-SIM	`similarity`	report_passages, similarity_checks, similarity_matches	3.10
FR-ATT	`attendance`	attendance_weeks	3.11
FR-EVL	`evaluations`	evaluations, evaluation_scores, company_evaluations	3.12
FR-WFL-02 – 07	`grades`	grades, grade_component_scores	3.13
FR-DSH	`dashboards`, `evaluations`	(queries), outcome_surveys	3.14
FR-EXP	`exports`	files	3.15
FR-CER	`certificates`	certificates	3.16
FR-NOT, FR-PWA-02	`notifications`, `jobs`	notifications, push_subscriptions, job_runs	3.17
FR-AUD	`core`	audit_log	3.18
NFR-SEC-04	`files`	files	3.19

10. Open Issues

Issue	Needed by	Plan
Wireframes: user walkthrough not yet done	Before Increment 1 ends	Walk through the clickable wireframes (Section 8.1) with at least one student and one faculty member (SRS 6.4.2, prototyping) and record the changes. Add the three screens listed as not yet drawn.
Example grade components, weights, and rubric criteria for the demo cycle	Increment 1	Seed example values (for example: company final evaluation 40%, weekly reports 30%, final report 30%) and confirm them with the course supervisor (SRS 6.4.5).
Skill vocabulary content (English and Arabic names, synonyms, resources)	Increment 1	Seed about 150 skills common in computing majors; extend while labelling the 20 CVs.
Labelled AI evaluation sets (20 CVs and profiles, 10 copied and 10 original reports)	Increment 2	Generate synthetic data now; a second person must label independently (SRS 6.4.2).
Calibration of `semantic_floor` and `semantic_ceiling` and the similarity thresholds	Increment 2 – 3	Tune with `ai-eval/evaluate.py`; record the chosen values and results in the test report.

Issue	Needed by	Plan
Project Gmail account, Hugging Face account and Space, Supabase project, Vercel project	Before Increment 1 ends	Create the free accounts; store the secrets as in Section 3.4.
Platform Admin account and demo organizations	Increment 1	The seed script creates the Platform Admin, 2 universities with 4 colleges, majors, the skill vocabulary, and sample companies; real Platform Admin accounts are created only by the provider (FR-AUT-02).